

INTERNSHIP PROJECT REPORT



Production Risk and Weather Analytics for Mining Operations and Accident Prediction

Submitted in the requirements for the award of the degree of

MASTER OF SCIENCE

IN

DATA SCIENCE

Submitted By

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Madras School of Economics



Internship Organisation: COAL INDIA LIMITED (ERP)

Internship Duration: 1 June 2026 – 31 July 2026

Under the Guidance of:

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Coal India Limited



CERTIFICATE

This is to certify that the project report entitled “**Production Risk and Weather Analytics for Mining Operation and Accident Prediction**” is a bona fide work carried out by **Hia Pal, student of Master of Science (Data Science), Madras School of Economics, Chennai**, during her industrial internship at Coal India Limited from **1 June 2026 to 31 July 2026**.

The project was developed in collaboration with Aryan Singh under the guidance of the ERP Department, Coal India Limited.

The work presented in this report fulfils the requirements of industrial training and demonstrates practical application of data science and machine learning principles, predictive modelling using ensemble methods including LightGBM, XGBoost, and Random Forest, feature engineering on real-world coal production and weather datasets, and the development of an interactive analytical dashboard and AI chat bot using LLM models for Coal India Limited's production risk assessment and weather impact quantification.

We wish her success in all future academic and professional endeavours.

Place: _____

Date: _____

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Shri Samir C. Sadhu
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STUDENTS DECLARATION

I undersigned Hia Pal a student of M.Sc. Data Science-MSE Business School, declare that summer internship project titled “Production Risk and Weather Analytics for Mining Operation and Accident Prediction” is a result of my own work and my indebtedness to other work publications, references, if any, have been duly acknowledged. If I/we are found guilty of copying any other report or published information and showing as my original work, I understand that I shall be liable and punishable by Institute, which may include ‘Fail’ in examination, ‘Repeat study & resubmission of the report’ or any other punishment that Institute may decide. Furthermore, I undertake that I will not use, disclose, share, reproduce, or distribute any data, information, or materials obtained during my internship outside the scope of this project without the prior written permission of Coal India Limited.

Date:

Place: CIL HQ, Kolkata

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Coal India Limited (CIL)

ACKNOWLEDGEMENT

I express my sincere gratitude to Coal India Limited, particularly the ERP Department, for providing me with the opportunity to undertake industrial training and gain practical exposure to real-world coal production data, weather impact analysis, and predictive modelling in an operational mining context, interactive dashboard development for production risk visualization, and the integration of an AI-powered conversational chatbot for query-driven insights

I would like to convey my heartfelt thanks to Mr. Samir Sadhu, General Manager (ERP), and Mr. Sayantan Pandit, Manager (ERP), for their invaluable guidance, continuous encouragement, and technical insights throughout the internship period.

I also extend my gratitude to the management and staff members of Coal India Limited who provided the necessary resources and support for successful completion of the project.

I am thankful to Madras School of Economics, Chennai, for offering an academic environment that encourages practical learning and professional development.

I would also like to acknowledge my fellow intern Aryan, with whom I collaborated during the development of the project. The collaborative effort significantly contributed to the successful implementation of various modules of the system.

Finally, I express my gratitude to my teachers, friends, and family members for their constant support and encouragement throughout the internship and project development process.

Abstract

Coal mining operations are greatly affected by weather conditions such as heavy rainfall, flooding, high temperatures, and strong winds. These weather events can reduce coal production, delay dispatch, and increase the risk of accidents. This project was carried out during an internship at Coal India Limited (CIL) with the aim of developing a data-driven system that helps predict weather-related production losses and identify accident-prone conditions before they occur.

The study used five years of weather data together with two years of actual production and offtake data collected from 47 mining areas across seven Coal India subsidiaries. Since only two years of operational data were available, synthetic data for three additional years was generated using a Random Forest-based approach to create a larger and more representative dataset. Several weather-related features, such as rainfall accumulation, waterlogging, consecutive rainy days, heat stress, and wind conditions, were created to better capture the effect of weather on mining operations.

Area-specific LightGBM machine learning models were then developed to predict daily production loss percentages. The models achieved good predictive performance, with an average R^2 of 0.926, an MAE of 1.86%, and an RMSE of 3.94%. A 15-day forecasting framework was also developed using weather forecasts to estimate future production losses and help managers identify high-risk areas in advance.

Along with production forecasting, a rule-based accident risk framework was designed to estimate weather-related safety risks. The system evaluates seven different hazards, including road slipperiness, lightning, ballast washout, wind hazards, water inrush, and slope instability, and combines them into a single accident risk score for each mining area.

Finally, all models were integrated into an interactive Streamlit dashboard with an AI-powered chatbot to provide easy access to forecasts, risk information, and operational insights. The project demonstrates how machine learning, weather analytics, and interactive visualization can support better planning, improve operational efficiency, and enhance safety in large-scale mining operations.

Keywords:

Coal India Limited, Machine Learning, LightGBM, Random Forest, Weather Analytics, Production Forecasting, Accident Risk Prediction, Streamlit Dashboard, Explainable AI, SHAP.

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Internship Overview

1. Scope of Work

Internship Overview

The internship involved participation in the design and implementation of a machine learning based production risk analytics system for Coal India Limited along with a risk.

Major areas covered included:

Production Risk Forecasting

- Data collection and preprocessing of real-world coal production and weather datasets
- Feature engineering including waterlogging index, consecutive rainy days, rainfall interaction terms, and weather productivity weights
- Synthetic data generation for FY2020–2023 to address data scarcity
- Predictive modelling using LightGBM (area-wise models) and Random Forest
- Anomaly detection using Isolation Forest to identify operational irregularities
- Walk-forward validation and model performance benchmarking
- Recursive 15-day forecasting of production loss percentage and expected production, per mining area
- Interactive dashboard for mine-level production loss forecasting and monitoring

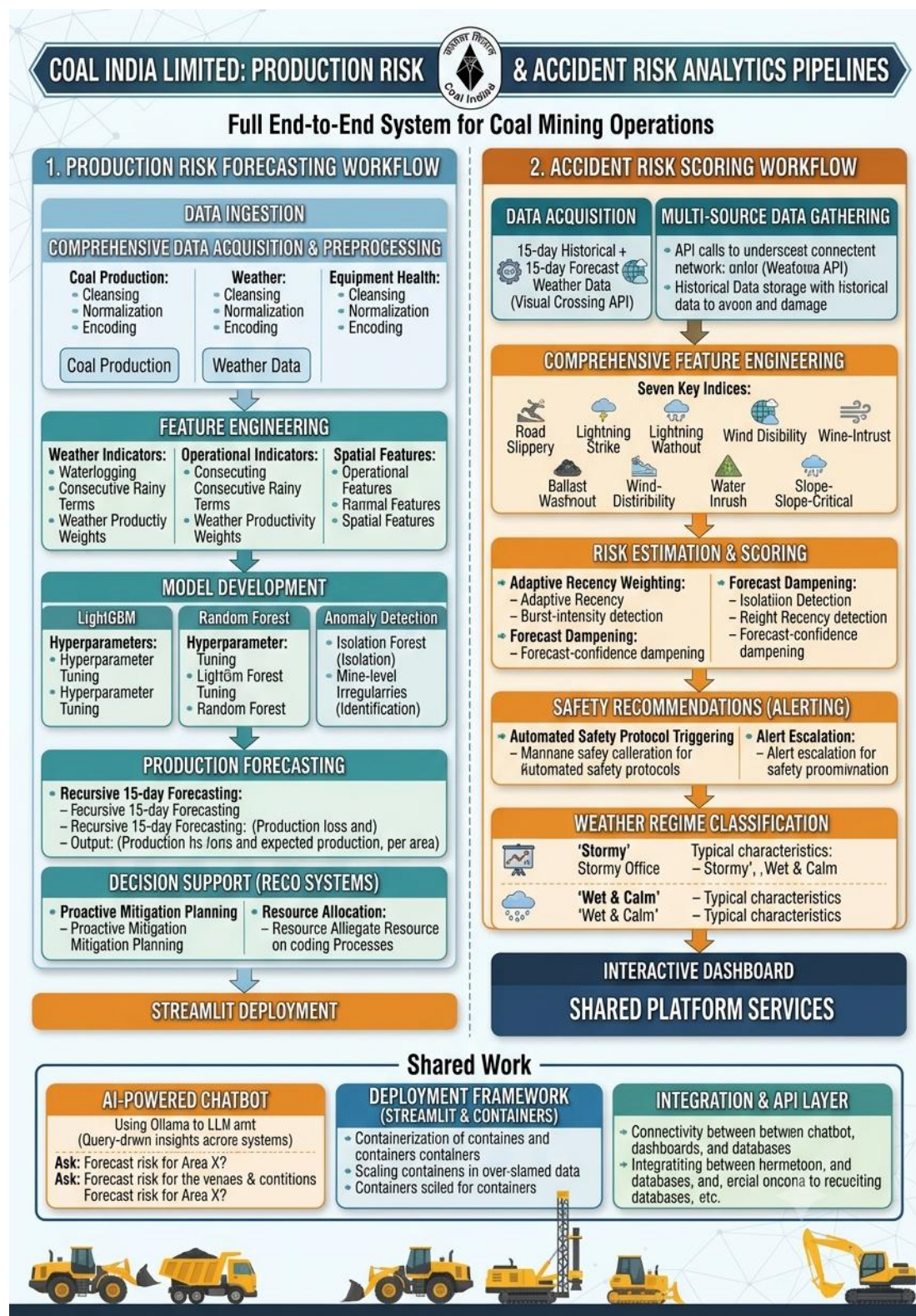
Accident Risk Scoring

- Fetching of 15-day historical + 15-day forecast weather data (Visual Crossing API) per mining area
- Development of a multi-index weather-accident risk scoring pipeline covering seven indices — Road Slippery, Lightning Strike, Ballast Washout, Wind-Dust Visibility, Wind-Structural, Water Inrush, and Slope-Critical
- Adaptive recency-weighted rainfall features, burst-intensity detection, and forecast-confidence dampening for robust risk estimation
- Composite risk score and weather regime classification (e.g. "Stormy," "Wet & Calm") per area
- Interactive dashboard for weather-driven accident risk visualization

Shared/Platform Work

- AI-powered chatbot using Ollama to run LLM integration for query-driven insights across both systems
- Deployment of the end-to-end analytics pipelines using Streamlit

2. Full Workflow



Project Introduction

1. Problem Statement

How weather disrupts production and creates deficits.

Currently our work focuses on weather-based issues in meeting daily or monthly target quotas rather than other logistical issues. CIL operates numerous mines which consist of both opencast and underground mines across whole India. Since India being a land of diverse climatic regions, different subsidiaries need to address the issues differently.

However Particularly during the monsoon season, adverse effects frequently go on to disrupt the mining activities which result in production and dispatch shortfalls. Along with the monsoon, one reason people tend to look over is high stress and temperature. Many of CIL mines are located in regions which receive only short bursts of rain sometimes and are only arid regions where temperatures may reach more than 40 degrees Celsius.

Beyond production and dispatch losses, adverse weather also poses a direct safety risk to mining operations and personnel. Heavy or prolonged rainfall can saturate haul roads and access routes, making them slippery and increasing the likelihood of vehicle skidding and slope failures, particularly in opencast mines with exposed benches and inclines. Intense rainfall events can also trigger ballast washout along rail and conveyor routes, and in extreme cases lead to water inrush into pits and underground workings, threatening both equipment and worker safety. Lightning and thunderstorm activity, common during the monsoon, raises the risk of strikes on personnel and infrastructure operating in open areas.

Since CIL is responsible for the majority of India's coal production, such disruptions can impact the daily or monthly quota which is given by the ministry.

2. Objectives of the study.

We have tried to take past weather data and production and dispatch data and tried to model a machine learning framework which will help us predict future discrepancies in coal production and offtake given weather forecasts. Our specific objectives remain to:

1. To examine the impact of weather conditions on coal production and offtake in Coal India Limited.
2. To identify weather patterns that contribute to production deficits and target shortfalls.
3. To develop machine learning models for predicting weather-induced production deficits.
4. To create an early warning framework for proactive production management.
5. To support quota recovery planning through data-driven deficit forecasting.
6. To recommend strategies for minimizing weather-related disruptions and improving target achievement.
7. To help predict accident prone conditions beforehand and increase in safety standards.

3. Scope — Areas across 7 subsidiaries

The model primarily focuses on analysing the impact of weather conditions on coal production performance within CIL during 5 financial years 2020-2025.

1. Geographical Coverage: The study covers 47 mining areas distributed across 7 Coal India subsidiaries:

- Bharat Coking Coal Limited (BCCL)
- Central Coalfields Limited (CCL)
- Eastern Coalfields Limited (ECL)
- Mahanadi Coalfields Limited (MCL)
- Northern Coalfields Limited (NCL)
- South Eastern Coalfields Limited (SECL)
- Western Coalfields Limited (WCL)

2. Data Coverage

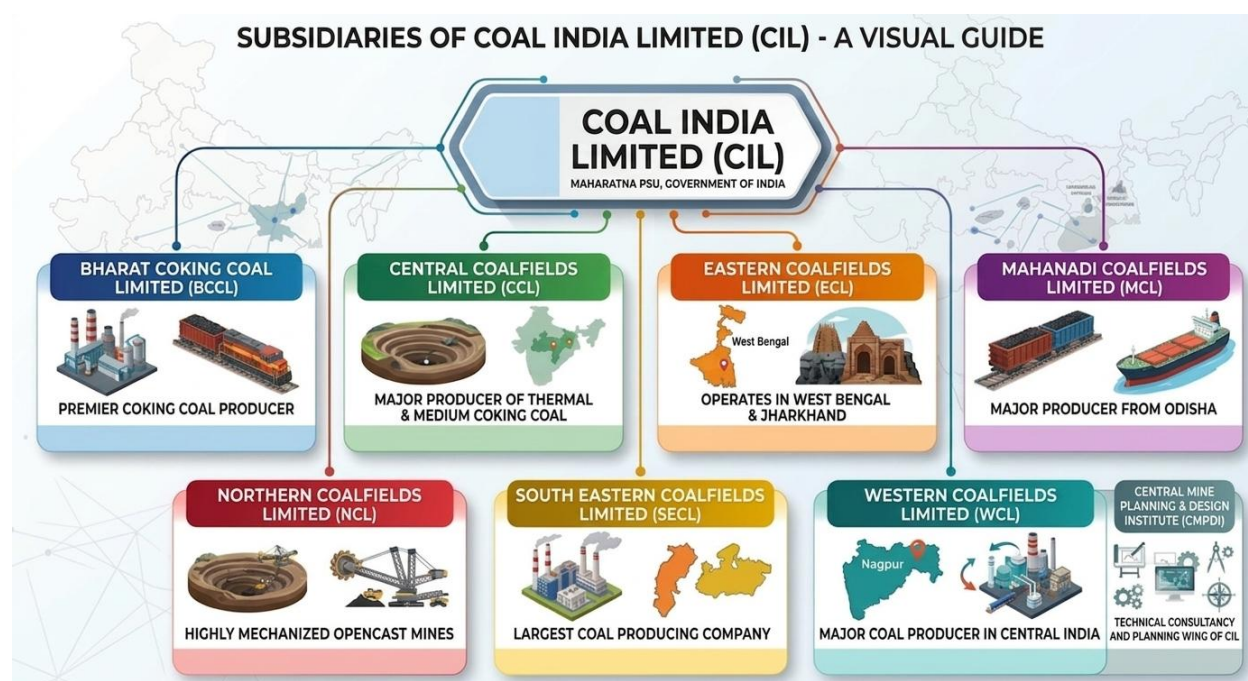
- Daily coal production and offtake records.
- Production targets and achievement levels.
- Feature engineered weather parameters
- Area level geographical information.

3. Analytical Scope

- Assessment of the relationship between weather conditions and production performance.
- Identification of weather-induced production deficits.
- Development of machine learning models to predict production shortfalls.
- Estimation of production recovery requirements following weather-related disruptions.
- Assessment of the relationship between weather conditions and safety/accident risk (road slipperiness, lightning, ballast washout, wind-driven hazards, water inrush, and slope stability).
- Development of a multi-index, area-wise weather-accident risk scoring framework using historical and forecast weather data.
- Estimation of near-term (15-day) accident risk outlook per mining area to support preventive safety planning

4. Limitations of Scope

- The study primarily examines weather-related operational impacts and does not explicitly model other factors such as equipment breakdowns, labour disruptions, geological constraints, regulatory interventions, or market-driven demand fluctuations.
- The accident risk indices are rainfall-, wind-, and weather-derived proxies; several indices (notably Water Inrush and Slope-Critical) lack subsurface hydrology, slope angle, and site-specific geotechnical data, and are therefore flagged with lower confidence.
- Findings are intended to support operational planning within the selected CIL subsidiaries and areas covered by the dataset.



Data Overview for Production Risk

1. Data Description

We have utilized a combination of weather, operation and synthetic datasets to analyze and predict the production and offtake deficits across the 7 subsidiaries in CIL.

2. Daily production and offtake record

Real Production and Offtake Data	FY2024-25
	FY2023-24
Weather Data	2020-2021
	2021-2022
	2022-2023
	2023-2024
	2024-2025

We have used these data records as the primary source for training the machine learning models for detection and production.

3. Weather Data

Daily weather observations were collected for each Area Headquarters covering FY2020–21 to FY 2024–25 using true area latitude and longitude using the Open-Meteo API which provides weather information though gridded meteorological data sets rather than individual weather stations. It used reanalysis products such as ERA5: combining satellite observation. Station measurements and numerical weather prediction models.

Raw Weather Variables used:

Raw Variable	Unit	Description
date	Date	Date of the weather observation.
area_code	-	Unique identifier for the mining area.
total_precipitation_mm	mm	Total precipitation (rainfall) received during the day.
precipitation_hours	Hours	Total number of hours during which precipitation occurred.
max_temp_c	°C	Maximum temperature recorded during the day.
min_temp_c	°C	Minimum temperature recorded during the day.
mean_temp_c	°C	Average temperature for the day.

max_wind_speed_kmh	km/h	Highest sustained wind speed during the day.
max_wind_gust_kmh	km/h	Maximum instantaneous wind gust recorded during the day.

4. Synthetic Data Generation using Random Forest:

Considering we only had real-time datasets of production and dispatch for FY 2023-24 and FY 2024-25 we have combined that with real-time 5 years weather data sets to generate synthetic data for FY 2020-2021, FY 2021-2022 and FY 2022-2023. The main reason being such a limited historical period is insufficient for capturing diverse weather conditions and also operational disruptions. The limited availability of operational data restricted the ability to capture a wide range of weather conditions, seasonal variations, and operational disruptions. Therefore, a synthetic data generation framework was developed to extend the historical record while preserving realistic weather-production relationships.

5. Weather Feature Engineering.

To get a holistic view of the different weather variables which affect production and dispatch we tried feature engineering. Feature engineering is the process of using domain knowledge and statistics to transform raw data into informative features that make machine learning models train faster and perform more accurately. It helps build better and robust predictive models.

Feature Engineered Variables:

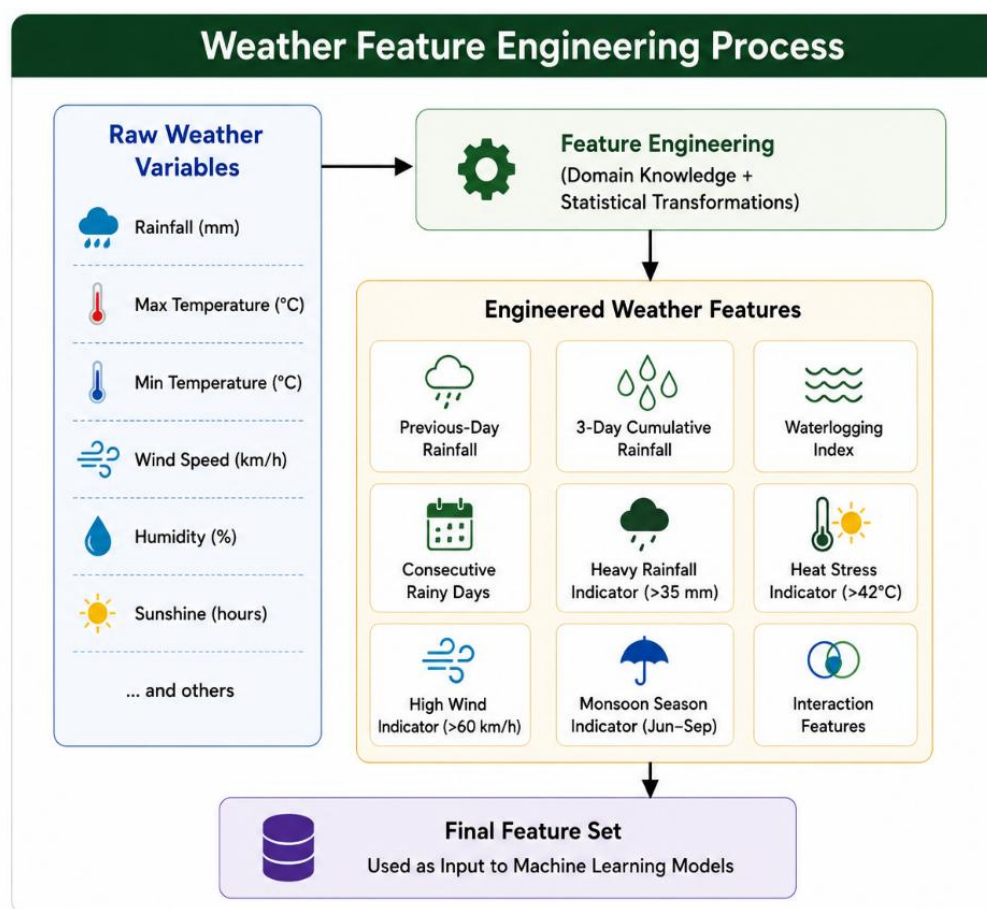
Variable	Formula / Definition
Previous-Day Rainfall	Rainfall observed on day (t-1)
Three-Day Cumulative Rainfall	Sum of rainfall on days (t), (t-1), and (t-2)
Waterlogging Index	Three-Day Cumulative Rainfall ÷ 50 (maximum value capped at 1)
Consecutive Rainy Days	Count of successive days with rainfall exceeding 0.5 mm
Heavy Rainfall Indicator	Binary variable equal to 1 when rainfall exceeds 35 mm
Heat Stress Indicator	Binary variable equal to 1 when maximum temperature exceeds 42°C
High Wind Indicator	Binary variable equal to 1 when maximum wind speed exceeds 60 km/h

Monsoon Season Indicator

Binary variable equal to 1 during June–September and 0 otherwise

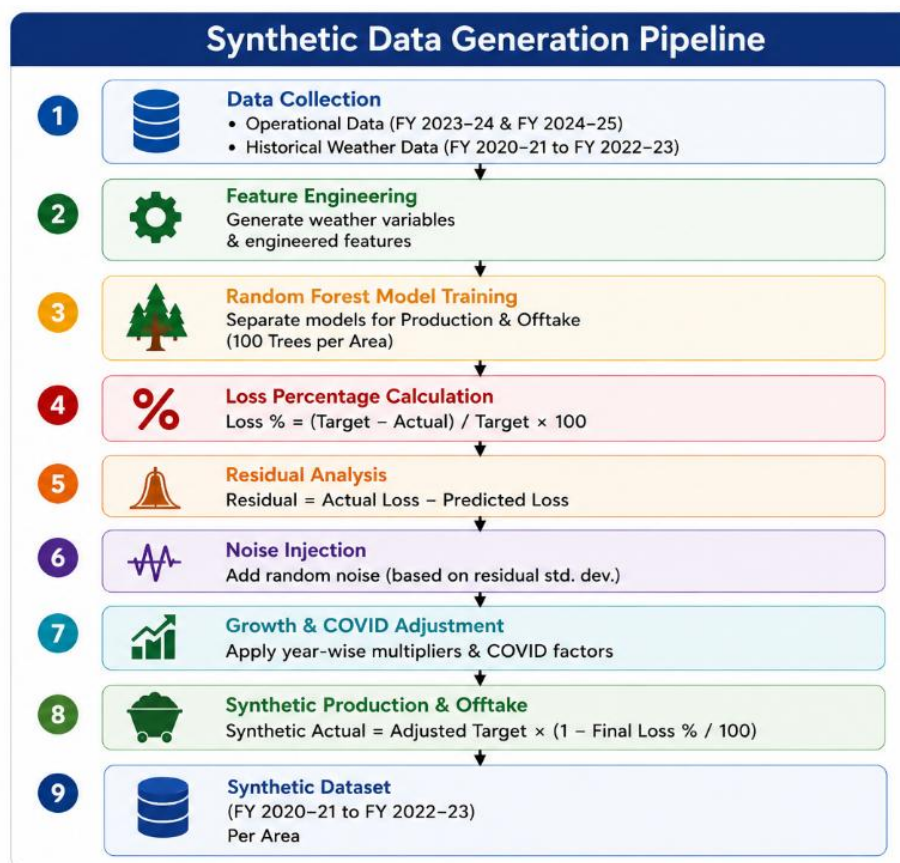
In addition to the above variables, several supplementary weather-derived features were also generated and evaluated during the exploratory analysis phase. Additional interaction variables, including Rainfall $\times$ Monsoon, Waterlogging $\times$ Monsoon, Heat Stress $\times$ Summer, and High Wind $\times$ Winter, were generated to capture the combined effects of weather conditions and seasonal patterns on mining operations.

These additional features were considered to capture more complex weather patterns and their potential impact on coal production and offtake performance.



6. Generation Methodology

Random forest Regression was selected because it has the ability to capture nonlinear relationships between the weather conditions and production losses. Separate Random Forest models comprising 100 decision trees were trained for each mining area to capture area-specific weather-production relationships. Daily production and offtake records for FY 2023–24 and FY2024–25 were combined with corresponding area-level weather data. Weather variables and engineered weather features were merged using Area Code and Date as matching keys.



Separate Random Forest models were developed for Production (PROD) and Offtake to capture the distinct impact of weather conditions on mining and evacuation activities.

For each observation, a production loss percentage was calculated as:

$$\text{Loss Percentage} = ((\text{Target} - \text{Actual}) / \text{Target}) \times 100$$

The loss percentage represents the extent to which actual production or offtake fell short of the planned targeted serves as a quantitative measure of operational disruption and enables the model to learn the relationship between weather conditions and production performance across different mining areas.

Model for Synthetic Dataset

One model was developed for each mining area, as every area exhibits a unique relationship between weather conditions and production losses. After training, the model generates a predicted loss percentage for each observation. This value represents the weather-induced production deficit expected under the observed weather conditions.

$$\text{PredictedLossPercentage} = f(\text{Weather Features}, \text{Engineered Features})$$

where $f(\cdot)$ represents the relationship between weather conditions and production loss learned by the Random Forest model.

The predicted loss percentage generated by the model was compared against the actual observed loss percentage to calculate the residual error.

$$\text{Residual} = \text{Actual Loss Percentage} - \text{Predicted Loss Percentage}$$

The residual represents the portion of production loss that could not be explained by weather variables alone. These unexplained variations may arise due to equipment breakdowns, labour availability, maintenance activities, transportation constraints, geological conditions, and other site-specific operational factors.

To preserve realistic operational variability within the synthetic dataset, the standard deviation of residuals was calculated separately for each area. During synthetic data generation, random noise based on the historical residual distribution was added to the model predictions.

$$\text{Final Loss Percentage} = \text{Predicted Loss Percentage} + \text{Random Noise}$$

This ensures that the synthetic dataset does not consist of perfectly smooth machine learning predictions and instead reflects the natural variability observed in real-world mining operations.

Historical weather data for FY 2020–21, FY 2021–22, and FY2022–23 were then passed through the trained area-specific Random Forest models to estimate the expected production loss percentage under the observed weather conditions for each day.

1. Growth and COVID Adjustments

To account for differences in historical production capacity, year-specific growth multipliers were incorporated into the synthetic data generation framework. Growth multipliers of 0.82, 0.85, and 0.91 were applied for FY 2020–21, FY 2021–22, and FY2022–23 respectively.

In addition, COVID-19 adjustment factors were introduced to reflect the operational disruptions experienced during the pandemic period. For FY 2021–22, adjustment factors of 0.55, 0.75, and 0.92 were applied during April–June, July–September, and October–March respectively. For FY 2022–23, an

adjustment factor of 0.82 was applied during April–May, while a factor of 1.00 was applied during the remaining months.

The adjusted daily production target was calculated as:

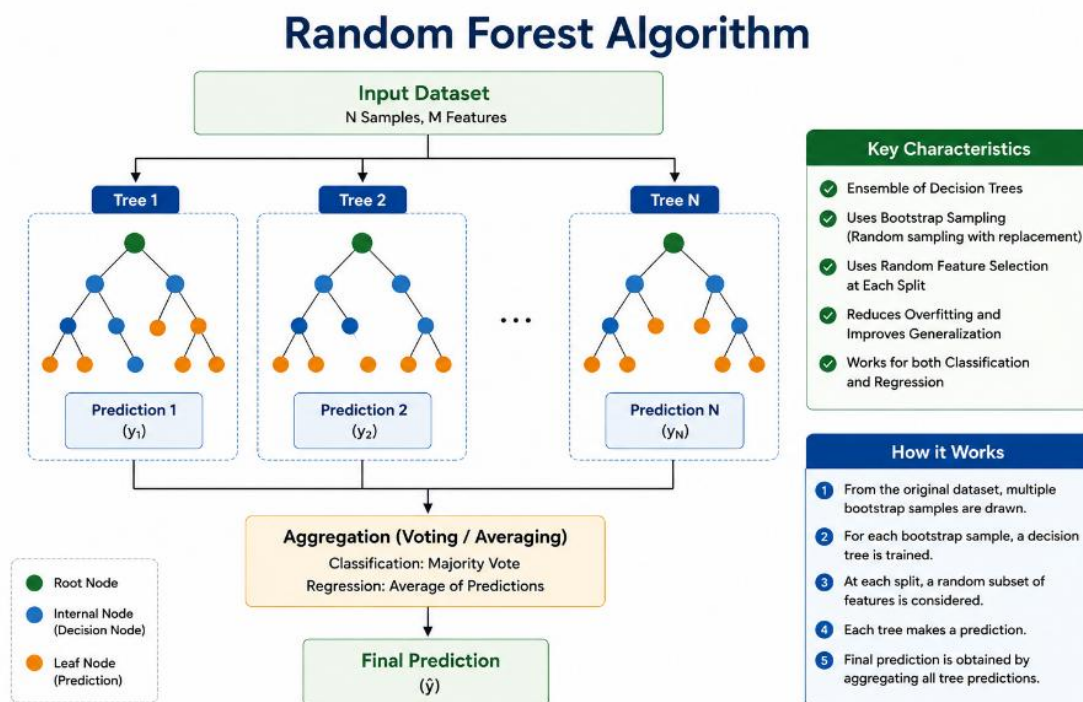
$$\text{Adjusted Daily Target} = \text{Baseline Daily Target} \times \text{Growth Multiplier} \times \text{COVID Adjustment Factor}$$

2. Generation of Synthetic Production and Offtake Values

Finally, synthetic production and offtake values were generated using:

$$\text{Synthetic Actual} = \text{Adjusted Daily Target} \times (1 - \text{Final Loss Percentage} / 100)$$

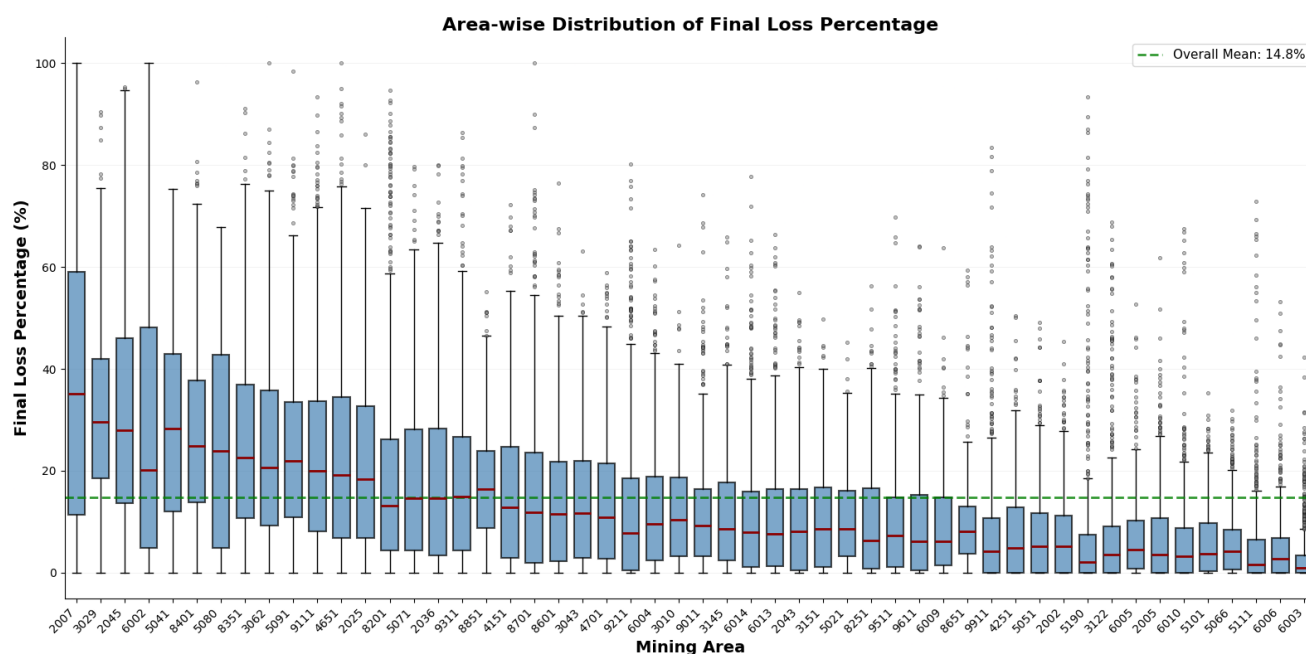
The above procedure was repeated for all eligible mining areas and all dates between FY 2020–21 and FY 2022–23. The generated synthetic production and offtake records were subsequently combined with the actual FY2023–24 and FY 2024–25 operational data to create a unified dataset for machine learning model development and analysis.



3. Validation

To ensure the synthetic dataset realistically represented historical mining operations, a series of validation checks were performed across four dimensions. Statistical distribution validation confirmed that the synthetic production and offtake values preserved key descriptive statistics observed in the actual FY2023–

24 and FY2024–25 datasets. Seasonal pattern validation verified that monthly and monsoon-period production trends were accurately reproduced, with appropriate disruption levels during high-rainfall months. Weather-production relationship validation confirmed that higher rainfall, prolonged wet periods, and elevated waterlogging conditions remained associated with greater production deficits, consistent with real operational behaviour. Finally, residual-based noise injection was applied during generation to preserve operational variability, preventing unrealistically smooth outputs. The validation results confirmed that the synthetic dataset successfully retained the statistical characteristics, seasonal patterns, and weather-driven relationships of the original data, making it suitable for extending the historical record and supporting machine learning model development for production deficit prediction and quota recovery planning.



Prediction Model

1. Overview

The prediction model was implemented as a per-area LightGBM regression pipeline designed to predict daily production loss percentage for each Coal India area headquarters. The pipeline was structured across nine sequential steps, taking as input the engineered weather features, lag-based production history, and offtake cross-signals, and producing area-level daily loss predictions, composite risk scores, and SHAP-based model explanations as outputs

2. Data Sources

Three primary data sources were used in the modelling pipeline. Real production and offtake data was loaded from the Fixed Production Dispatch Dataset, with records filtered separately for production and offtake material types. Synthetic production and offtake data covering FY2020–23, generated using the Random Forest framework described in the previous section, was used to augment the training set and address the limited availability of historical real data. Daily weather observations for FY2023–24 and FY2024–25 provided the weather input to the model. Areas with codes 5190, 6003, 8351, 4151, and 4251 were excluded from modelling due to insufficient or anomalous data coverage, resulting in 47 areas being included in the final model.

3. Production Loss Calculation

The target variable, Production Loss Percentage, was calculated for each area and date as:

$$\text{Production Loss \%} = \frac{\text{Target} - \text{Actual}}{\text{Target}} \times 100$$

Values were clipped to the range [0, 100] to eliminate negative loss arising from overperformance and to cap extreme outliers. For synthetic records, the Final Loss Percentage column generated during synthetic data creation was used directly as the target variable.

4. Feature Engineering

A total of 72 features were constructed across three groups.

4.1 Weather Features

The weather feature set consisted of variables describing rainfall, temperature, wind, seasonality, and their interactions. These included rainfall aggregates, waterlogging indicators, consecutive rainy days, binary weather flags, seasonal and calendar variables, and interaction terms designed to capture compound weather effects. In addition, two derived variables Rainfall Intensity, calculated as rainfall divided by precipitation hours, and Temperature Range, calculated as the difference between maximum and minimum daily temperature were engineered to better represent rainfall severity and daily thermal variation

4.2 Lag and Rolling Features

Historical production information was transformed into lag and rolling statistics to capture temporal dependencies and production trends. These features included production volume lags, production momentum, production loss lags, rolling averages and variability measures of production loss, together with first- and higher-order changes in production loss to represent short-term production dynamics.

4.3 Offtake Cross-Signal Features

Offtake-derived features were engineered to capture demand-side and logistics-related influences on production. These features included historical offtake loss patterns, rolling summary statistics, offtake momentum, and variables describing the relationship between production and offtake, enabling the model to account for operational imbalances between coal production and dispatch.

ENGINEERED FEATURES (TOTAL: 72)			
4.1 WEATHER FEATURES (TOTAL: 36)			
No.	Sub-Category	No. of Features	Feature Name
1	A. RAINFALL FEATURES (7)	7	1. Rainfall (mm)
2			2. Previous-Day Rainfall
3			3. 3-Day Cumulative Rainfall
4			4. 7-Day Cumulative Rainfall
5			5. 15-Day Cumulative Rainfall
6			6. 7-Day EMA of Rainfall
7			7. Rainfall Intensity (Rainfall / Precipitation Hours)
8	B. WATERLOGGING & PERSISTENCE (2)	2	8. Waterlogging Index
9	9. Consecutive Rainy Days		
10	C. TEMPERATURE FEATURES (4)	4	10. Maximum Temperature (°C)
11			11. Minimum Temperature (°C)
12			12. Mean Temperature (°C)
13			13. Temperature Range (Max – Min) (°C)
14	D. WIND & OTHER METEOROLOGICAL (2)	2	14. Maximum Wind Speed (km/h)
15	15. Precipitation Hours		
16	E. CALENDAR & SEASONAL FEATURES (12)	12	16. Day of Week
17			17. Day of Month
18			18. Day of Year
19			19. Week of Year
20			20. Quarter
21			21. Sunday Indicator (1=Sunday, 0=Otherwise)
22			22. Quarter-End Month Indicator (1=Yes, 0=No)
23			23. Monsoon Season Indicator (1=Monsoon, 0=Other)
24			24. Summer Indicator (1=Summer, 0=Other)
25			25. Monsoon Indicator (1=Monsoon, 0=Other)
26			26. Post-Monsoon Indicator (1=Post-Monsoon, 0=Other)
27			27. Winter Indicator (1=Winter, 0=Other)
28	F. BINARY WEATHER FLAGS (4)	4	28. Rain Flag (1=Rain, 0=No Rain)
29			29. Heavy Rain Flag (1=Heavy Rain, 0=No)
30			30. Heat Stress Flag (1=Heat Stress, 0=No)
31			31. High Wind Flag (1=High Wind, 0=No)
32	G. INTERACTION FEATURES (5)	5	32. Rainfall × Monsoon
33			33. Rainfall × Temperature
34			34. Waterlogging × Monsoon
35			35. Heat Stress × Summer
36			36. High Wind × Winter

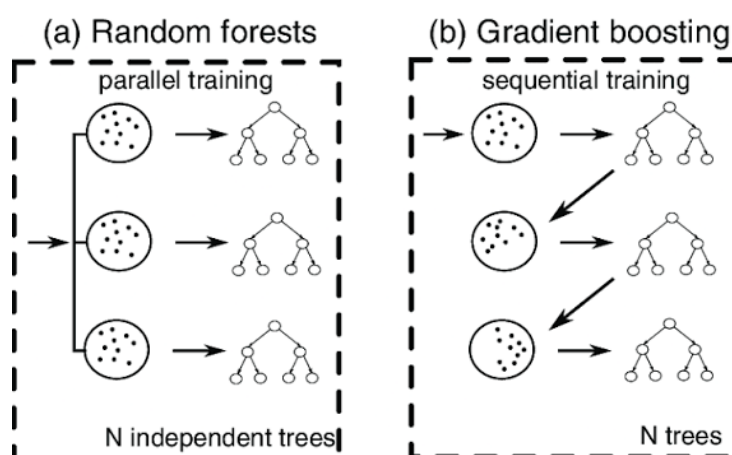
4.2 LAG AND ROLLING FEATURES (TOTAL: 21)			
No.	Sub-Category	No. of Features	Feature Name
37	A. PRODUCTION VOLUME LAGS (6)	6	37. Production Lag (1 Day)
38			38. Production Lag (2 Days)
39			39. Production Lag (3 Days)
40			40. Production Lag (7 Days)
41			41. Production Lag (14 Days)
42			42. Production Lag (30 Days)
43	B. PRODUCTION MOMENTUM (3)	3	43. Production Momentum (3 Days)
44			44. Production Momentum (7 Days)
45			45. Production Momentum (14 Days)
46	C. PRODUCTION LOSS LAGS (5)	5	46. Production Loss Lag (1 Day)
47			47. Production Loss Lag (2 Days)
48			48. Production Loss Lag (3 Days)
49			49. Production Loss Lag (7 Days)
50			50. Production Loss Lag (14 Days)
51	D. ROLLING STATISTICS OF PRODUCTION LOSS (3)	3	51. 7-Day Average Production Loss
52			52. 30-Day Average Production Loss
53			53. 7-Day Standard Deviation of Production Loss
54	E. LOSS CHANGE FEATURES (4)	4	54. 1-Day Loss Change
55			55. 3-Day Loss Change
56			56. 7-Day Loss Change
57			57. Loss Acceleration (Second Difference of Production Loss)

4.3 OFFTAKE CROSS-SIGNAL FEATURES (TOTAL: 15)			
No.	Sub-Category	No. of Features	Feature Name
58 59 60 61 62	A. OFFTAKE LOSS LAGS (5)	5	58. Offtake Loss Lag (1 Day) 59. Offtake Loss Lag (2 Days) 60. Offtake Loss Lag (3 Days) 61. Offtake Loss Lag (7 Days) 62. Offtake Loss Lag (14 Days)
63 64 65	B. ROLLING STATISTICS OF OFFTAKE LOSS (3)	3	63. 7-Day Average Offtake Loss 64. 30-Day Average Offtake Loss 65. 7-Day Standard Deviation of Offtake Loss
66 67 68	C. OFFTAKE MOMENTUM (3)	3	66. Offtake Momentum (3 Days) 67. Offtake Momentum (7 Days) 68. Offtake Momentum (14 Days)
69 70 71 72	D. PRODUCTION-OFFTAKE RELATIONSHIP FEATURES (4)	4	69. Production-to-Offtake Ratio 70. Production-Offtake Loss Gap 71. 7-Day Average Production-to-Offtake Ratio 72. 7-Day Average Production-Offtake Loss Gap

SUMMARY OF FEATURES		
Feature Group	No. of Features	Percentage of Total
Weather Features	36	50.00%
Lag and Rolling Features	21	29.17%
Offtake Cross-Signal Features	15	20.83%
TOTAL FEATURES	72	100.00%

5. Model Architecture

A LightGBM gradient boosting regressor was trained independently for each area, allowing the model to capture area-specific weather-production relationships rather than fitting a single global model. Features were standardised using a Standard Scaler prior to training. Real data rows were assigned a sample weight of 3.0 relative to synthetic rows to reflect their higher operational fidelity. Each area model was trained on the combined synthetic FY2020–23 and real FY2023 dataset and evaluated on held-out real FY2024 data. Areas with fewer than 30 training rows or fewer than 5 test rows were excluded from evaluation. The hyperparameters used were 300 estimators, maximum depth of 6, learning rate of 0.05, 31 leaves, minimum child samples of 10, subsample ratio of 0.8, column sample ratio of 0.8, and L1 and L2 regularisation values of 0.1 each.



6. Risk Scoring

A composite risk score in the range [0, 100] was computed for each area-date record by combining the model's predicted loss with supporting weather signals:

$$\text{Risk Score} = 0.35 \cdot L + 0.15 \cdot R + 0.12 \cdot W + 0.10 \cdot C + 0.08 \cdot H + 0.20 \cdot O$$

Where L is the predicted loss percentage, R is the three-day rainfall sum scaled by 1.8, W is the waterlogging index scaled to 100, C is consecutive rainy days scaled by 4, H is the heat stress component computed as $\max(0, (\text{Max_Temp} - 35) \times 3)$, and O is the offtake loss percentage scaled by 0.5. All components were clipped to $[0, 100]$ before weighting. Records were categorised as Low (0–24), Moderate (25–49), High (50–74), or Severe (75–100) based on the computed score.

RISK SCORE CALCULATION – EXAMPLE

$$\text{Risk Score} = 0.35 \cdot L + 0.15 \cdot R + 0.12 \cdot W + 0.10 \cdot C + 0.08 \cdot H + 0.20 \cdot O$$

Where,

L = Predicted production loss (%)

R = Three-day rainfall sum scaled by 1.8

W = Waterlogging index scaled to 100

C = Consecutive rainy days scaled by 4

H = Heat stress component = $\max(0, (\text{Max_Temp} - 35) \times 3)$

O = Offtake loss (%) scaled by 0.5

1. INPUT VALUES AND SCALED COMPONENTS

Component	Raw Input Value (Example Day)	Scaling Applied	Scaled Value (Clipped to [0, 100])
L Predicted production loss (%)	45%	$L = 45$	45.00
R Three-day rainfall sum (mm)	20 mm	$R = 1.8 \times 20$	36.00
W Waterlogging index	0.40	$W = 100 \times 0.40$	40.00
C Consecutive rainy days	4 days	$C = 4 \times 4$	16.00
H Maximum temperature (°C)	39 °C	$H = \max(0, (39 - 35) \times 3)$ $= 4 \times 3$	12.00
O Offtake loss (%)	30%	$O = 0.5 \times 30$	15.00

Note: All scaled values are within [0, 100]; therefore, no clipping was required in this example.

2. RISK SCORE CALCULATION

$$\begin{aligned} \text{Risk Score} &= 0.35(45.00) + 0.15(36.00) + 0.12(40.00) + 0.10(16.00) + 0.08(12.00) + 0.20(15.00) \\ &= 15.75 + 5.40 + 4.80 + 1.60 + 0.96 + 3.00 \\ &= \mathbf{31.51} \end{aligned}$$

Risk Score = 31.51

3. RISK CATEGORY ASSIGNMENT

Risk Score Range	0 – 24	25 – 49	50 – 74	75 – 100
Risk Category	Low	Moderate	High	Severe

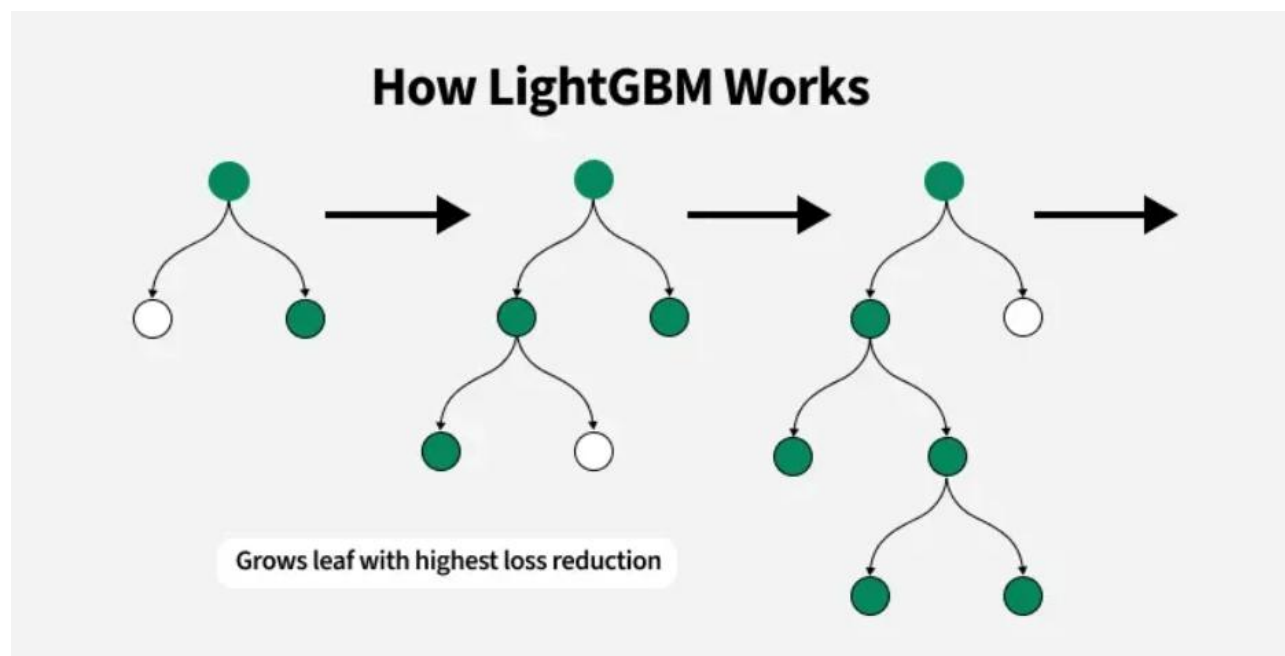
Since the calculated Risk Score is **31.51**, the area for this day is classified as **MODERATE RISK**.

7. Model Explainability

SHAP values were computed for each trained area model on a sample of up to 30 FY2024 test records. For each sample, the top five features increasing predicted loss and the top five features decreasing predicted loss were recorded, providing area-level interpretability of model outputs and enabling operationally meaningful explanations alongside the numerical forecasts.

8. Why LightGBM?

LightGBM was selected as the primary modelling algorithm for the following reasons.



Performance on Tabular Data

LightGBM consistently delivers state-of-the-art performance on structured tabular datasets, which is the nature of the coal production and weather data used in this study. It has demonstrated superior accuracy compared to traditional machine learning algorithms such as Random Forest and standard gradient boosting on similar regression tasks.

Handling of Large Feature Sets

With 79 engineered features spanning weather variables, lag terms, rolling statistics, and offtake cross-signals, the dataset required a model capable of efficiently handling high-dimensional input. LightGBM uses a histogram-based learning algorithm that bins continuous features, significantly reducing memory usage and training time without sacrificing predictive accuracy.

Leaf-wise Tree Growth

Unlike conventional decision tree algorithms that grow trees level-wise, LightGBM grows trees leaf-wise, selecting the leaf with the highest potential loss reduction at each step. This results in deeper, more accurate trees with fewer iterations, which was particularly beneficial given the per-area training approach where some areas had limited training samples.

Built-in Support for Sample Weights

The pipeline assigned a sample weight of 3.0 to real operational records to prioritise learning from actual observed data over synthetic records. LightGBM natively supports sample-weighted training, making this weighting straightforward to implement.

Compatibility with SHAP Explainability

LightGBM integrates directly with the SHAP TreeExplainer, enabling efficient and exact computation of feature attributions for each prediction. This was essential for the explainability component of the pipeline, which required interpretable, area-level explanations of model outputs.

Validation Against Alternatives

Random Forest and XGBoost were also evaluated during the modelling phase. While Random Forest provided reasonable baseline performance, LightGBM achieved lower MAE and higher R^2 on the FY2024 test set across the majority of areas, confirming its suitability as the final model.

9. Results

9.1 Model Performance Evaluation

```
Performance Summary:
Areas trained: 47
Avg R2: 0.926
Avg MAE: 1.86%
Avg RMSE: 3.94%
Features: 72
```

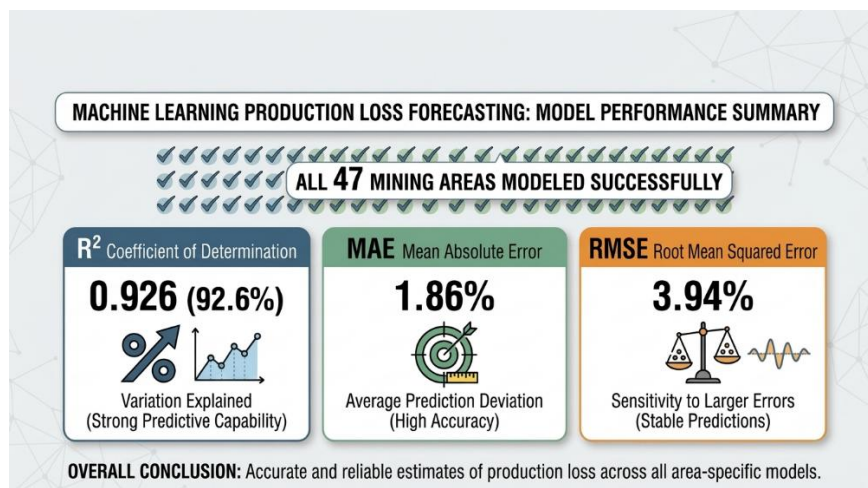
The model was successfully trained for all 47 mining areas, allowing area-specific relationships between weather conditions and production loss to be captured.

The average coefficient of determination ($R^2 = 0.926$) indicates that approximately 92.6% of the variation in production loss is explained by the model. This demonstrates a strong predictive capability and suggests that the selected weather, operational, and historical production variables effectively capture the factors influencing production performance.

The Mean Absolute Error (MAE) of 1.86% indicates that, on average, the predicted production loss differs from the actual loss by only 1.86 percentage points.

Similarly, the Root Mean Squared Error (RMSE) of 3.94% shows that even larger prediction errors remain relatively small. Since RMSE penalizes larger errors more heavily than MAE, the low RMSE suggests that the model produces stable predictions without frequent large deviations.

Overall, these evaluation metrics demonstrate that the proposed machine learning framework provides accurate and reliable estimates of production loss across different mining areas.



9.2 Risk Classification Results

After predicting production loss, the model converts the predicted loss into operational risk categories to assist mine managers in prioritizing intervention.

A total of 16,927 daily records were evaluated.

✓ Risk scores: 16,927 records

Risk Distribution:

Low : 13,412 (79.2%)
Moderate : 2,978 (17.6%)
High : 527 (3.1%)
Severe : 10 (0.1%)

he majority (79.2%) of observations fall within the Low Risk category, indicating that mining operations are generally able to meet production targets under normal operating conditions.

Approximately 17.6% of observations are classified as Moderate Risk, suggesting that adverse weather or operational conditions may reduce production but are still manageable with corrective planning.

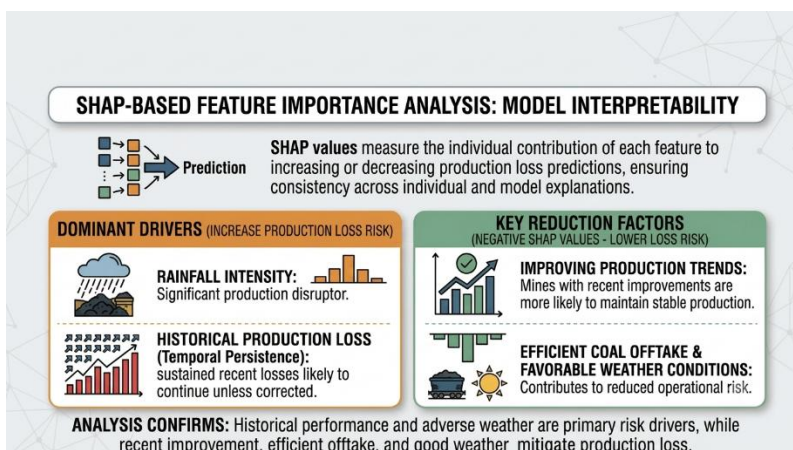
Only 3.1% of observations fall into the High Risk category, while 0.1% are classified as Severe Risk. These represent rare but significant events such as prolonged heavy rainfall or extreme operational disruptions that may require immediate management intervention.

The skewed distribution towards the Low Risk category is expected because severe weather events occur relatively infrequently compared to normal operating conditions.

9.3 SHAP-Based Feature Importance Analysis

To improve model interpretability, SHAP (SHapley Additive exPlanations) analysis was performed. SHAP values quantify the contribution of each feature toward increasing or decreasing the predicted production loss.

Unlike traditional feature importance measures, SHAP explains the influence of individual variables on each prediction while preserving consistency across the model.



The results indicate that historical production loss variables dominate the prediction, suggesting that production performance exhibits strong temporal persistence. If a mine has experienced sustained production losses over recent days, similar conditions are likely to continue unless corrective actions are implemented.

Rainfall intensity also emerges as an important predictor, confirming that adverse weather conditions significantly disrupt coal production.

Negative SHAP values indicate that these variables push the prediction toward **lower production loss**. The analysis suggests that mines demonstrating improving production trends during recent days are more likely to maintain stable production. Efficient coal offtake and favorable weather conditions also contribute to reduced operational risk.

=====

SHAP SUMMARY - TOP FEATURES

=====

- TOP FEATURES THAT INCREASE LOSS:
 - Loss_Change_7: +9.4631 (count: 116)
 - Loss_Change_3: +7.0205 (count: 92)
 - Rainfall_Intensity: +6.4614 (count: 1)
 - Loss_7day_Avg: +4.1845 (count: 175)
 - Loss_Lag_1: +3.5657 (count: 135)
 - Loss_Change_1: +3.1019 (count: 93)
 - Loss_Lag_7: +1.1977 (count: 190)
 - Loss_Lag_3: +1.1260 (count: 142)
 - Season_Summer: +1.0654 (count: 40)
 - Loss_30day_Avg: +1.0579 (count: 98)
- TOP FEATURES THAT DECREASE LOSS:
 - Loss_Lag_1: -3.4210 (count: 316)
 - Loss_7day_Avg: -3.1996 (count: 251)
 - Temp_Range: -3.1719 (count: 11)
 - Loss_Change_7: -2.6312 (count: 304)
 - Loss_Change_3: -2.1252 (count: 310)
 - Loss_Change_1: -1.6977 (count: 270)
 - Offtake_Lag_1: -1.5469 (count: 13)
 - Loss_Lag_3: -1.5015 (count: 191)
 - Min_Temp_C: -1.2021 (count: 16)
 - Loss_Lag_7: -1.0883 (count: 170)

Forecast Model

1. Overview

The prediction model developed in the previous section estimates the Production Loss Percentage for historical observations where both weather conditions and operational data are already available. While this is useful for evaluating model performance, it cannot support operational planning because the actual production values for future dates are unknown.

To enable proactive decision-making, a forecasting framework was developed to estimate future production losses before they occur. Instead of relying on historical weather observations, the forecasting model uses daily weather forecasts obtained from the Open-Meteo Forecast API together with the trained area-specific LightGBM models. This enables Coal India to identify mining areas that are likely to experience production disruptions several days in advance, allowing preventive measures to be implemented before production is affected.

The forecasting framework generates daily predictions for the next fifteen days for every mining area. These forecasts subsequently form the basis for the operational Risk Score, allowing management to prioritise high-risk areas and allocate resources more effectively.

2. Data Sources

The forecasting framework relies on daily weather forecasts obtained from the Open-Meteo Forecast API. The API provides location-specific weather forecasts based on the geographical coordinates of each mining area. Latitude and longitude corresponding to every mining area are first retrieved from the area mapping dataset, after which separate weather forecasts are downloaded for each location.

3. Daily Target Calculation

For each area and each day in June, the predicted daily production loss percentage was generated by passing the combined weather and lag feature vector through the trained LightGBM model. Predictions were clipped to the range [0, 100]. The predicted daily production and weather productivity weight were then derived as

$$\text{Predicted Production} = \text{Daily Target} \times \left(1 - \frac{\text{Predicted Loss \%}}{100}\right)$$

The monthly production target was calculated as:

$$\text{Monthly Target} = \text{Daily Target} \times \text{Working Days}$$

where working days in June 2026 totalled 26,

4. Feature Engineering

The same weather feature engineering pipeline used during model training was applied to both the historical and forecast weather datasets, constructing rainfall aggregates, waterlogging index, consecutive rainy days, binary weather flags, calendar features, seasonal dummies, and interaction terms. Where engineered columns were already present in the input files they were retained directly. Missing columns were filled with zero or sensible defaults to ensure a consistent 79-feature input. The canonical feature ordering was loaded from the Feature Manifest file saved during training to guarantee exact alignment with the trained models.

5. Model Architecture

Per-area LightGBM models were loaded from disk and each model's stored feature list was verified against the canonical Feature Manifest. A rolling state dictionary was maintained for each area during forecast generation, tracking the running history of predicted loss, predicted production, and offtake loss values. From this state, all 22 lag and rolling features were computed dynamically for each forecast day.

Since no real production history was available for the period immediately preceding June 2026, a lag bridge was constructed by first generating predictions for March 2026 using historical weather data, and using the last 30 days of those predictions to initialise the lag state for each area. For areas where no trained model was available or where a feature list mismatch was detected, a rule-based fallback was applied, scaling the historical average loss upward proportionally based on rainfall and waterlogging thresholds of 10, 20, and 40 mm.

6. Risk Scoring

Monthly deficit alerts were computed for each area. The deficit was calculated as:

$$\text{Deficit} = \text{Monthly Target} - \text{Total Predicted Production}$$

where total predicted production was the sum of June 1–22 historical period predictions and June 23–30 forecast period predictions. Areas were classified into four risk levels: High Risk if a deficit was projected and average predicted loss exceeded 20 percent, Medium Risk if a deficit was projected and average loss exceeded 10 percent, Low Risk if a deficit was projected or peak predicted loss exceeded 30 percent, and On Track otherwise. Risk distributions were aggregated at the subsidiary level, reporting total projected deficit, average loss percentage, and counts of areas in each risk category across the seven subsidiaries.

8. Forecast Validation

The forecasting framework was validated by comparing forecasted and observed Production Loss Percentage over a 15-day validation period for each of the 47 mining areas. Performance was evaluated using MAE, RMSE, and R^2 . The model achieved an average MAE of 3.06% and average RMSE of 3.91%, demonstrating reliable short-term forecasting performance comparable to the training results (MAE = 1.86%, RMSE = 3.94%).

The R^2 metric was reported only for areas with sufficient variation in production loss, as it becomes unreliable when the target variance is very low. Only four mining areas (2045, 3145, 5071, and 8251)

recorded an MAE greater than 10%, suggesting the need for further model refinement. Overall, the results indicate that the proposed framework provides accurate and reliable short-term production loss forecasts across the majority of mining areas.

```
Areas validated : 47  
Avg MAE       : 3.06%  
Avg RMSE      : 3.91%
```


Data Overview for Accident Risk Prediction

1. Data Description

The accident risk framework uses two primary data sources: (i) daily weather data obtained from the Visual Crossing Weather API and (ii) road and soil properties used in the Road Slippery hazard index. Unlike the production forecasting model, this framework is entirely rule-based and therefore does not require historical model training. Instead, risk scores are computed directly from observed and forecast weather conditions using predefined hazard-specific rules.

2. Weather Data

Daily weather data were collected for each mining area using the Visual Crossing Weather API based on the latitude and longitude of each area. Geographic coordinates were converted from Degrees-Minutes-Seconds (DMS) to decimal degrees where required, and areas with missing or unreliable coordinates were excluded from the analysis. For each mining area, weather data consisted of 15 days of historical observations and 15 days of forecast data, with each record labelled as either Historical or Forecast.

- Maximum, minimum and mean temperature (°C)
- Relative humidity (%)
- Dew point temperature (°C)
- Cloud cover (%)
- Atmospheric pressure (hPa)
- Solar radiation (W/m² and MJ/m²)
- Daily precipitation (mm)
- Probability of precipitation (%)
- Wind speed and wind gust (km/h)
- Visibility (km)
- UV Index
- Severe weather risk score
- Weather conditions and descriptions

Two additional weather indicators were derived from the reported weather conditions. A Thunderstorm Flag identifies the occurrence of thunderstorms using weather icons and condition descriptions, while a Severe Weather Flag identifies potentially hazardous weather using severe weather keywords and the API's severe weather risk score. In addition, a Heat Index was calculated using the Rothfus regression to represent apparent temperature under hot and humid conditions.

Areas with codes 5190, 6003, 8351, 4151, and 4251 were excluded, consistent with the exclusion list used in the production risk pipeline, due to insufficient or unreliable coordinate/data coverage.

3. Road and Soil Data

Road and soil information was incorporated for the Road Slippery hazard index. Where available, Liquid Limit (LL), Plastic Limit (PL) and soil classification were used to estimate soil moisture behaviour. For

road segments with missing soil information, representative values based on standard Unified Soil Classification System (USCS) soil classes were assigned, with a general default used when no classification was available.

4. Feature Engineering

The weather variables were transformed into a set of engineered features representing rainfall persistence, atmospheric instability, surface moisture and environmental conditions. These engineered variables formed the inputs to the seven hazard indices and included rainfall accumulation, rainfall intensity, consecutive rainy days, pressure change, dew point depression, moisture indices, and other hazard-specific indicators. Each feature was calculated separately for every mining area before being used in the rule-based hazard classification framework.

Since several mining hazards depend on cumulative weather conditions rather than individual daily observations, the raw weather variables were transformed into a set of engineered features representing moisture accumulation, rainfall persistence, atmospheric instability and environmental conditions. These features formed the inputs to the hazard-specific risk indices.

The principal engineered features include:

- Fast Bucket: Represents short-term surface moisture resulting from recent rainfall.
- Slow Bucket: Represents long-term ground saturation by accumulating moisture over an extended period.
- Liquidity Index (LI): Estimates soil plasticity and haul-road slipperiness using moisture conditions together with soil properties.
- Rainfall Burst: Measures extreme short-duration rainfall events.
- Consecutive Rainy Days: Counts the number of successive wet days.
- Wet–Dry Cycles: Represents repeated wetting and drying of the road surface.
- Pressure Drop: Indicates rapidly developing storm systems.
- Dew Point Depression: Used as an indicator of fog formation and reduced visibility.
- Surge Ratio: Measures sudden increases in water availability relative to existing ground saturation.

Accident Risk Modelling

1. Overview

The accident risk framework is a rule-based scoring system developed to estimate weather-related accident risk across mining areas. Unlike the production risk model, which uses a machine learning model to make predictions, this framework combines engineered weather features with domain knowledge to evaluate different types of operational hazards. Instead of relying only on rainfall totals, the framework first models how moisture builds up and remains in the mining environment over time, and then combines this information with other weather conditions such as wind, pressure, humidity and visibility. These variables are used to calculate seven independent hazard indices, each representing a different accident mechanism. The individual hazard scores are then combined to produce a single composite accident risk score for every area and day, providing an overall measure of weather-related operational risk.

2. Data Sources

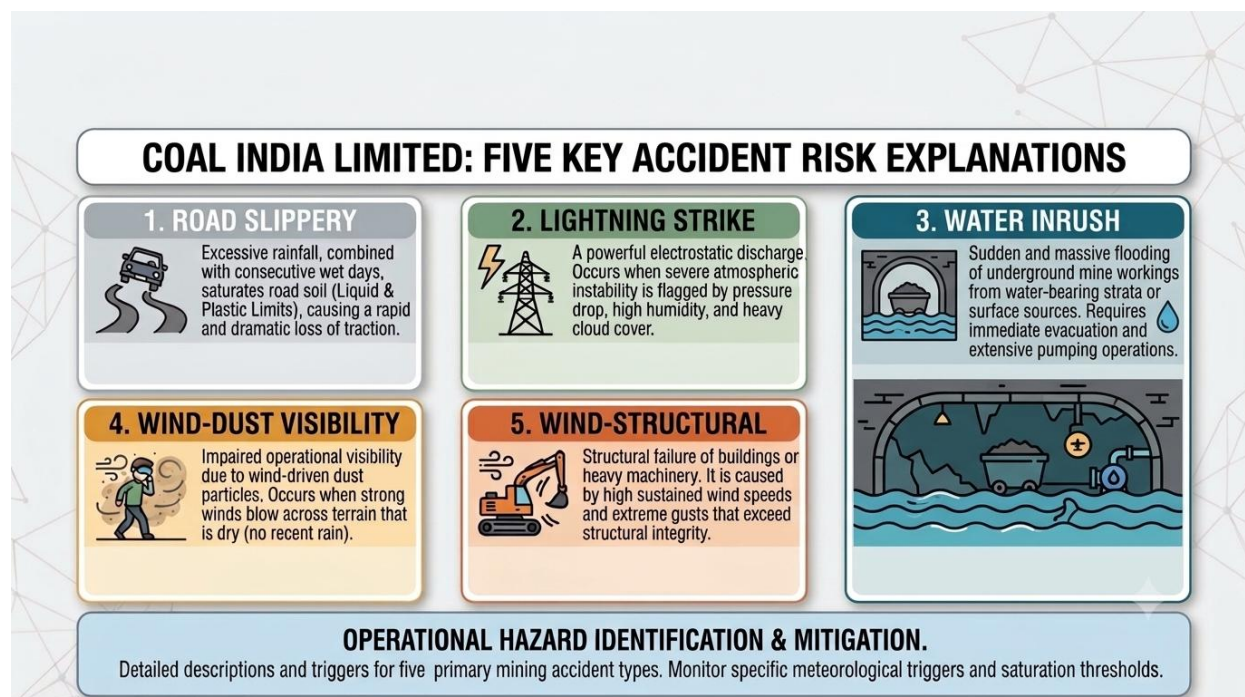
The model consumes the 30-day Historical + Forecast weather file described above, together with the optional road/soil material dataset. All seven indices are computed per Area Code and Date, across both the historical and forecast portions of the timeline; however, area-level risk summaries and rankings are computed on the Forecast window only, since that reflects the operationally relevant look-ahead period, while the Historical rows are retained in the full output for audit and back testing purposes.

3. The Seven Risk Indices

The accident risk framework evaluates seven independent hazard indices, with each index representing a different weather-related accident mechanism encountered in mining operations. Instead of combining all weather variables into a single model, each index uses only the weather conditions that are directly related to that particular hazard. This makes the framework easier to interpret and allows individual risks to be monitored separately. Every index produces a categorical risk level (Low, Moderate, High or Severe) together with a numerical score ranging from 0 to 3. These scores are later combined to calculate the overall composite accident risk score.

Each hazard index produces two outputs: a categorical risk level and a numerical score. The categorical risk level is classified into four categories: **Low**, **Moderate**, **High**, and **Severe**, while the numerical score ranges from **0 to 3**, where 0 represents Low risk and 3 represents Severe risk. Using a common scoring scale allows all seven indices to be compared directly and provides a consistent basis for calculating the overall composite accident risk score described in the next section.

The following subsections describe the methodology used to calculate each of the seven hazard indices, including the weather variables used, the reasoning behind their selection, and the rules used to classify the final risk level.



4. Composite Risk Score

The seven hazard indices are combined to produce a single composite risk score that represents the overall weather-related accident risk for each mining area on each day. Each hazard index generates a numerical score between 0 and 3, corresponding to the categories Low, Moderate, High, and Severe. The composite score is calculated by averaging the seven index scores and scaling the result to a range of 0–100.

To ensure that critical hazards are appropriately reflected in the final score, the following operational escalation rules are applied:

- If two or more hazard indices are classified as Severe, 10 points are added to the composite score (capped at 100).
- If three or more hazard indices are classified as Severe, the composite score is automatically raised to at least 75 (Severe).
- If either the Road Slippery or Slope-Critical index is classified as Severe, or if a Severe Weather Flag is present, the composite score is raised to at least 50 (High).

The final composite score is interpreted using the following categories:

- Low: 0–24
- Moderate: 25–49
- High: 50–74
- Severe: 75–100

5. Risk Classification Methodology

Each hazard index is computed using a set of weather-derived variables and engineered features specific to the corresponding accident mechanism. Rather than relying on a common classification rule, each index applies hazard-specific decision thresholds and, where appropriate, additional escalation conditions to capture the effects of extreme or persistent weather events. The resulting hazard score is subsequently used

in the computation of the composite risk score. The classification methodology for each hazard index is summarized below.

5.1 Road Slippery

- **Input variables:** Liquidity Index (LI), Consecutive Rainy Days, Wet–Dry Cycles.
- **Decision rule:** The Liquidity Index is calculated from the surface moisture conditions and compared against predefined thresholds. The score is further adjusted when repeated wet–dry cycles indicate deteriorating road conditions.

5.2 Lightning Strike

- **Input variables:** Thunderstorm Indicator, Severe Weather Flag, Pressure Drop, Relative Humidity, Cloud Cover and Solar Radiation.
- **Decision rule:** The hazard score is determined from atmospheric instability indicators, with additional escalation applied during severe convective weather.

5.3 Wind–Dust Visibility

- **Input variables:** Wind Speed, Visibility, Dew Point Depression and Fast Bucket.
- **Decision rule:** The hazard score is assigned using visibility degradation caused by either wind-driven dust or moisture-related fog conditions.

5.4 Wind–Structural

- **Input variables:** Wind Gust, Sustained Wind Speed and Pressure Drop.
- **Decision rule:** Structural risk is evaluated primarily from effective wind loading, with stronger gusts producing progressively higher hazard scores.

5.5 Water Inrush

- **Input variables:** Slow Bucket, Fast Bucket and Surge Ratio.
- **Decision rule:** The classification is based on long-term ground saturation together with sudden increases in surface water availability.

5.6 Slope-Critical

- **Input variables:** Slow Bucket, Consecutive Rainy Days and Extreme Rainfall Burst.
- **Decision rule:** The hazard score is determined from cumulative ground saturation and rainfall persistence, with escalation during extreme precipitation events.

Visualizations - Streamlit

1. Why Streamlit?

Streamlit was chosen to build the dashboard because it lets us turn python code straight into a working web app without having to build a separate frontend. It is Open Source, Free, and Easy to use.

Data models and machine learning were already written in Python, Streamlit's python libs let us plug that into user interface. Since this dashboard is data first, integrating it in viewable format was eased.

- **Fast to build** charts, filters, and KPIs running with simple coding.
- **Native Python integration:** works seamlessly with pandas, NumPy, scikit-learn models, and plotting libraries like Plotly.

Easy to run: The whole dashboard launches with a single command (streamlit run app.py), with zero maintenance costs.

Easy to install:

```
import streamlit as st
# Display an image from a local file path or URL
st.image("my_chart.png", caption="Production Loss Forecast Map")
```

Using the command:

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\merged_project> streamlit run app.py
2026-07-22 13:57:38.080 Uvicorn server started on ::8501

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://172.17.73.254:8501
External URL: http://210.212.31.91:8501
```

2. Drawbacks of Streamlit

While Streamlit was the right choice for building this project quickly, it does come with some limitations:

- **Re-run execution model:** Streamlit re-runs the entire script from top to bottom on almost every user interaction (a click, a dropdown change, etc.), which makes the dashboard feel slower than a traditional web app.
- **Limited UI customization:** because Streamlit generates its own HTML/CSS, a fully custom, pixel-perfect design is harder than with a dedicated front-end framework.
- **Not ideal for very large numbers of concurrent users:** Streamlit is best suited for internal tools and small-to-medium user bases rather than public, high-traffic consumer applications.

- Session state can get complex: as an application grows (multiple pages, a chatbot with conversation memory, working state, etc.), manually managing what is remembered across reruns requires careful design.
- Fewer native mobile-specific optimizations, since Streamlit is primarily built for desktop/browser dashboard use.



3 Scale of the Platform

It covers seven subsidiaries and roughly fifty-two area-level operations, each with its own historical weather records, production figures, and a dedicated forecasting model. This scale is one of the reasons the dashboard is organized into distinct pages rather than a single view: an executive needs a company-wide summary, while an area manager needs to be able to drill down into just their own operation without going through fifty other areas' data.

4 Tech Stack

At a high level, the project sits on top of a Python data and machine-learning stack: pandas for data handling, Plotly for the interactive charts seen throughout the dashboard, and a set of trained forecasting models

producing the predictions surfaced on the Loss Forecast and other pages. On top of this analytical foundation, Streamlit provides the web interface for the dashboard, while a separate, purpose-built conversational pipeline (detailed in Part B) power the chatbot. Both components read from the same underlying datasets, which is what keeps the numbers a user sees in the dashboard and the numbers the chatbot mentions in sync with one another.

5 Report Structure

The rest of this report is divided into two halves. The first half explains the Dashboard, what it shows, how it is organized, and what value it delivers to an operations user, described at a conceptual level without going into the underlying machine learning models. The second half explains the AI Copilot (chatbot), how it understands questions, retrieves information, and generates grounded answers, described with more technical depth since the chatbot's internal pipeline is central to understanding how it avoids giving misleading or made-up answers.

Visualizations - Dashboard

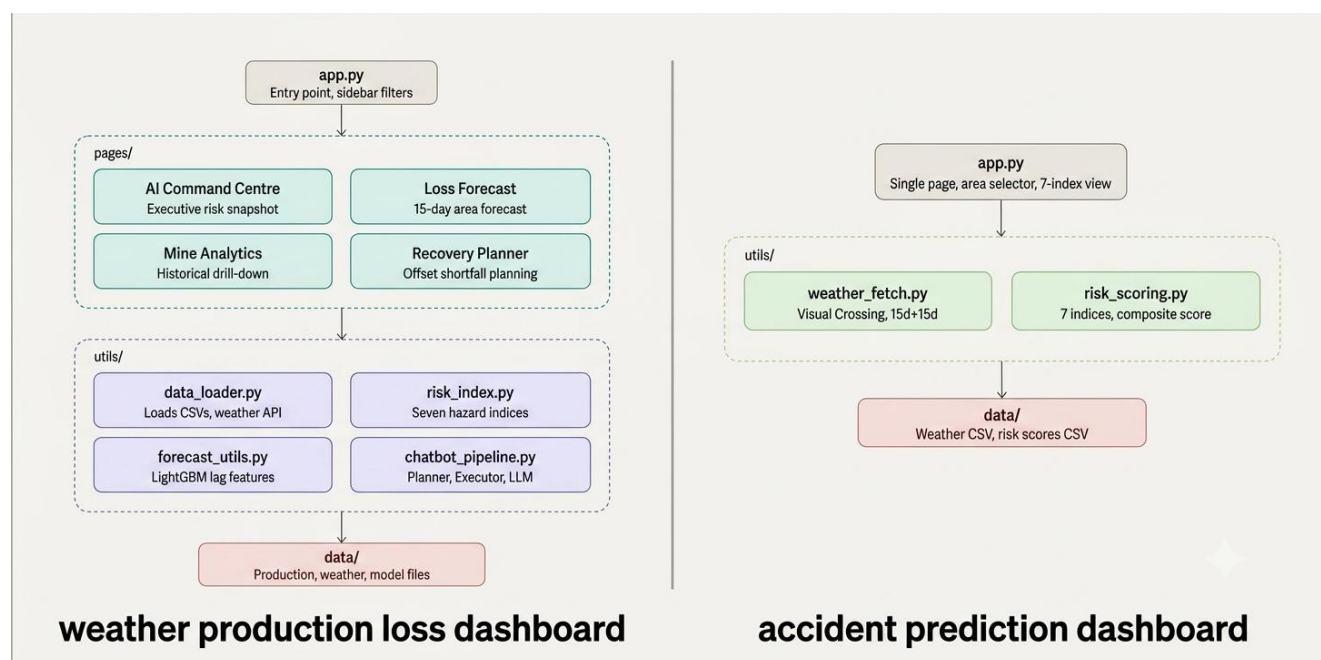
1. Purpose of the Dashboard

The dashboard is the visual heart of the platform. It is designed to help operations managers and executives quickly understand how much coal each area is producing, how weather, particularly rainfall, is affecting or is expected to affect production, which areas are at the highest risk, and what recovery actions could help offset expected losses. It removes the need to manually compare production spreadsheets with weather reports.

The dashboard is designed for different types of users. Area managers can monitor their own mines, subsidiary coordinators can compare performance across areas, and executives can view an overall company summary. To make the system easy to use, the dashboard is divided into separate pages instead of one long page. Each page follows the same layout and uses common sidebar filters for subsidiary, area, and fiscal year, allowing users to move between pages without having to learn a new interface.

The dashboard combines three types of information in one place: historical production data, historical and forecast weather data, and production and risk forecasts. This allows users to easily see how weather conditions relate to production. For example, they can identify whether a drop in production occurred during a period of heavy rainfall and whether similar conditions are expected in the coming days.

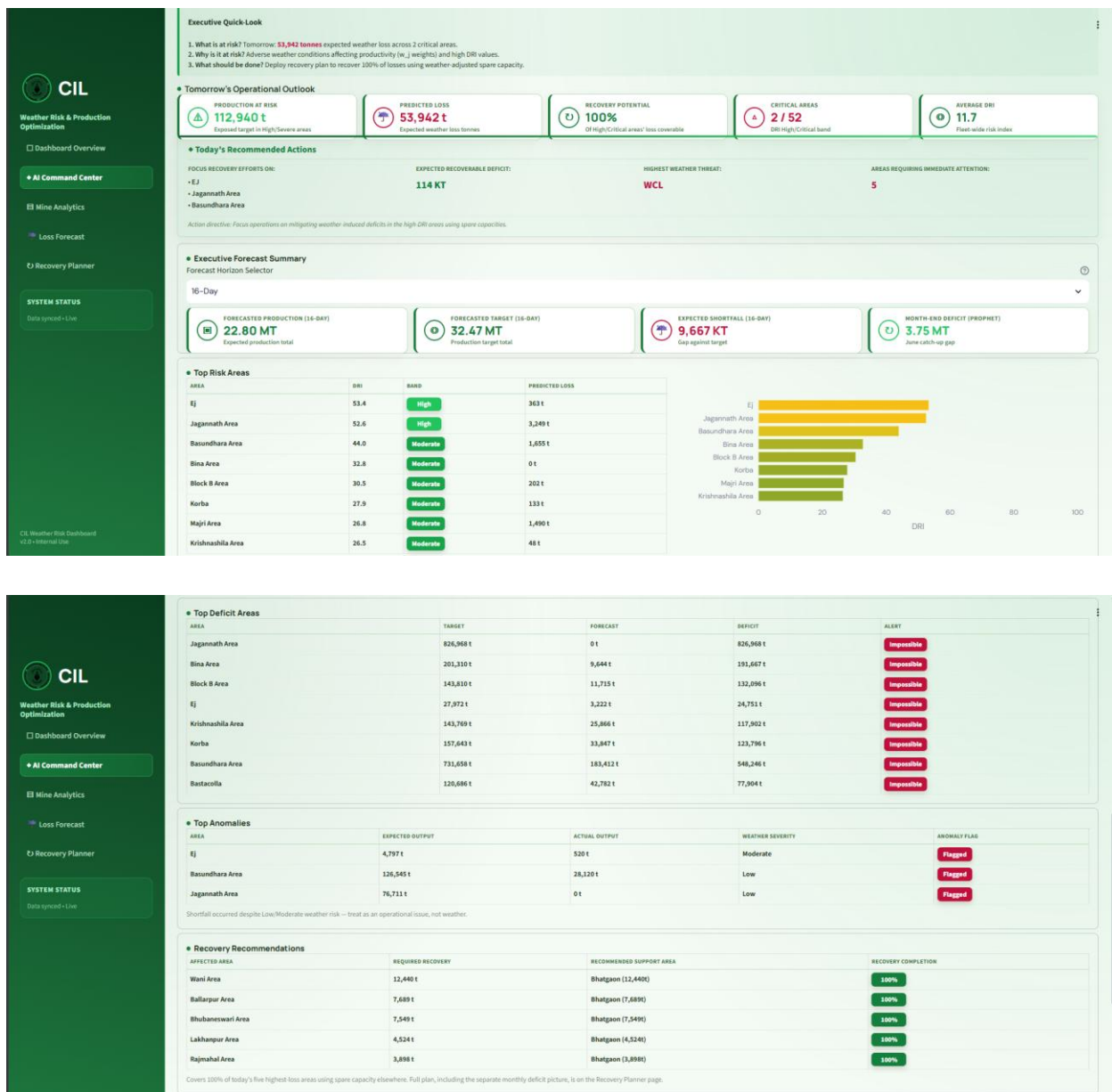
2. Flowchart



Dashboard Screen Shots

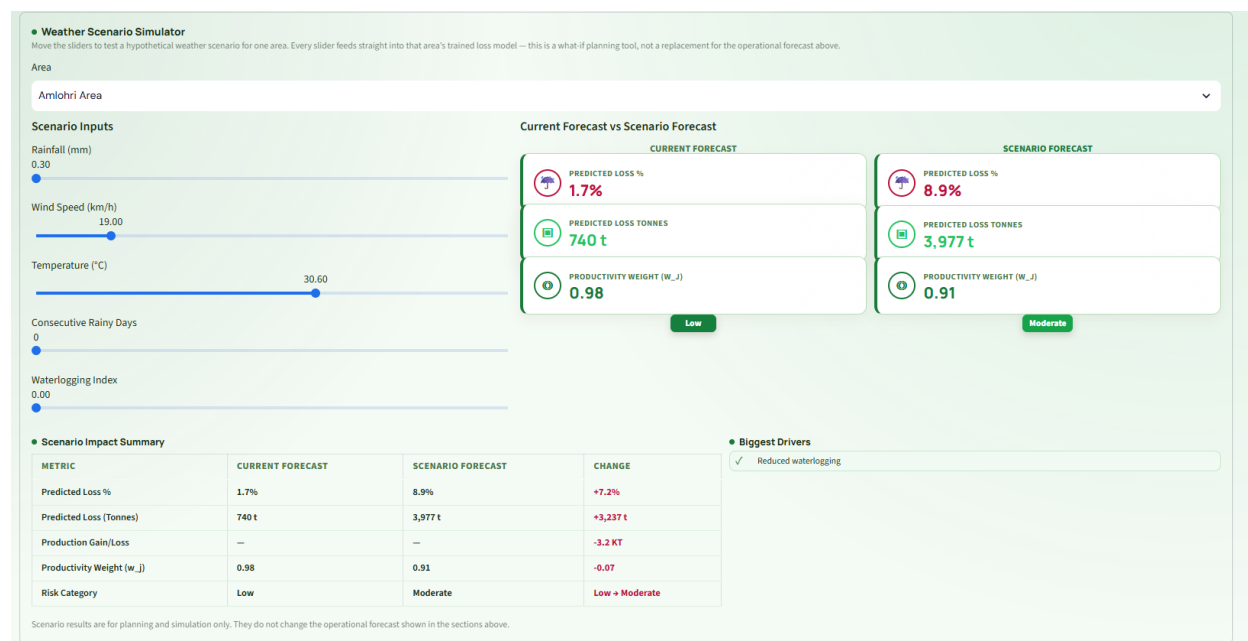
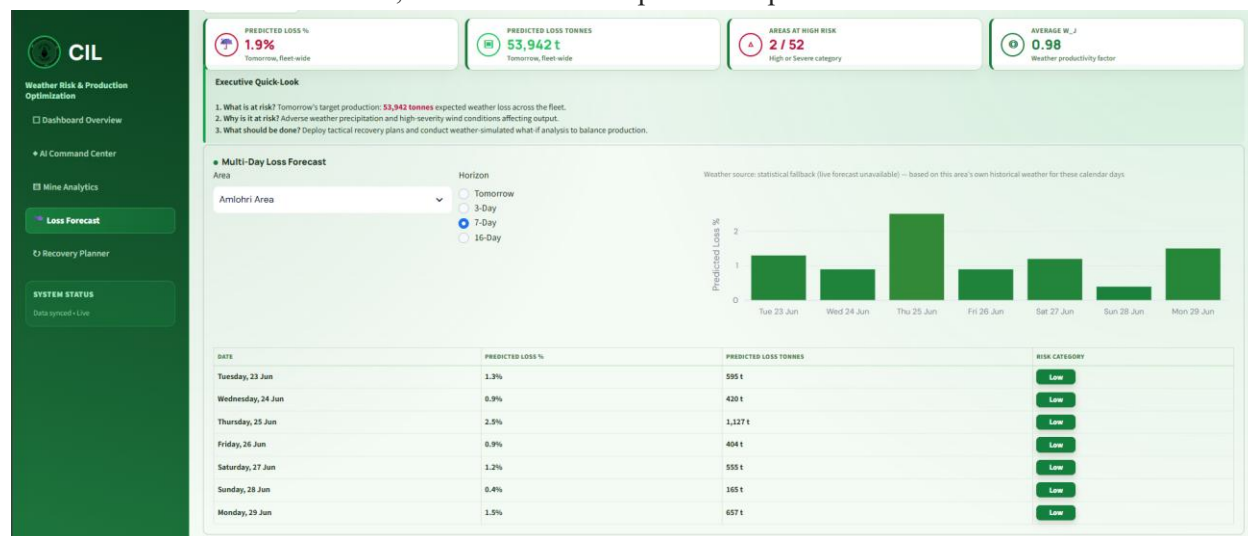
1. AI Command Centre

This is a one-screen overview meant to be understood at a glance. It brings together the most important indicators for the whole company or a selected subsidiary: which areas are at the highest risk, an overall Digital Risk Index, weather-driven alerts, and headline production numbers. The idea is that a senior stakeholder should be able to open this single page and immediately know 'where the problems are today' without navigating anywhere else.



2. Loss Forecast

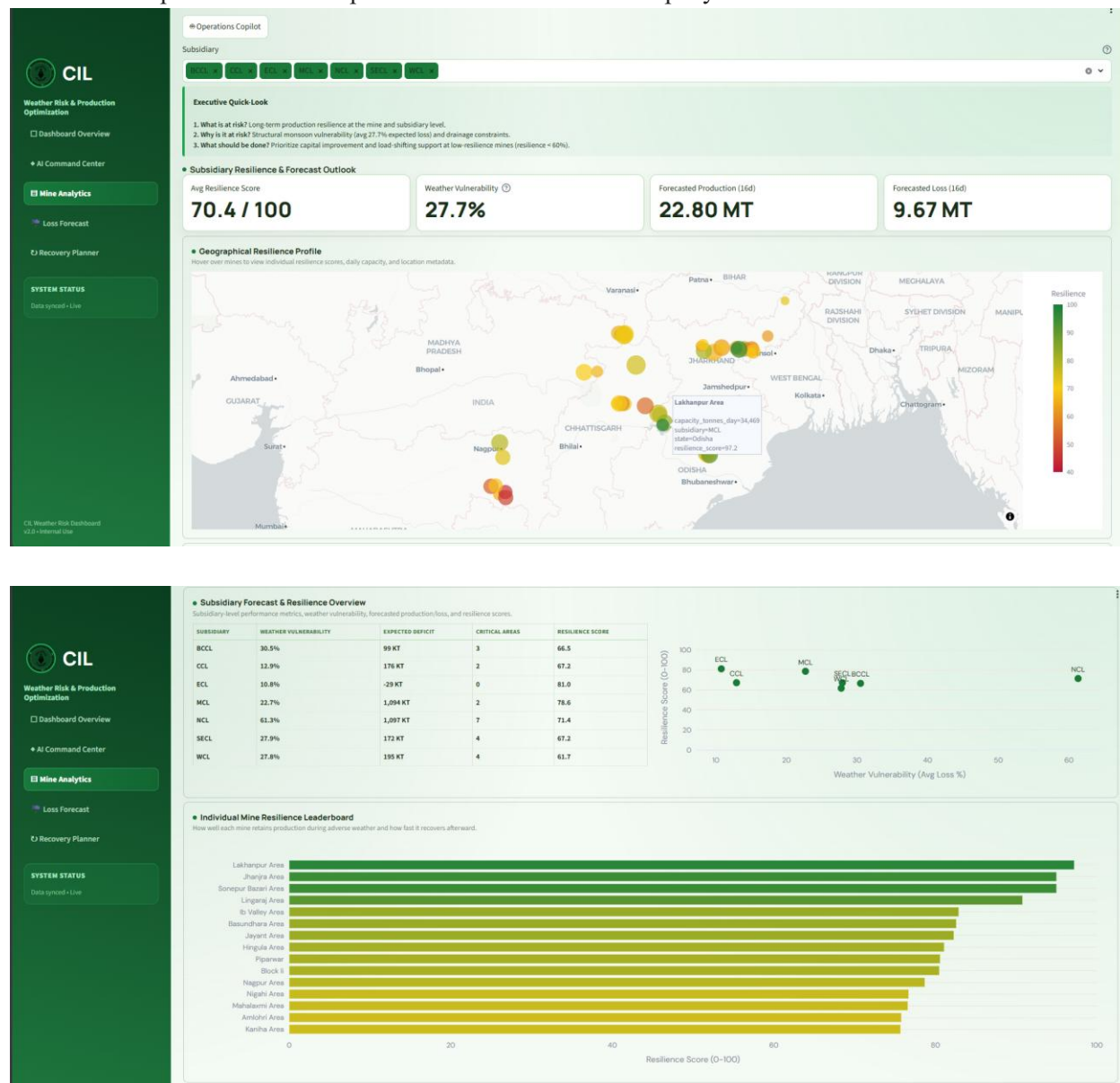
This page focuses on decision support at the area level. It shows how much production loss is expected in the near future due to weather risk and presents this in a way that helps a manager decide where to focus attention. Rather than a single point number, forecasts are shown with a confidence range, giving the user a sense of how reliable the prediction is rather than treating it as an exact guarantee. With help of open-meteo API and mine coordinates, accurate weather reports were possible.



The Sliders can be shifted, since the data is dynamic, this scenario is pushed in a scrapper that feeds the data and compares with the trained prediction models, based on which new forecast numbers are generated.

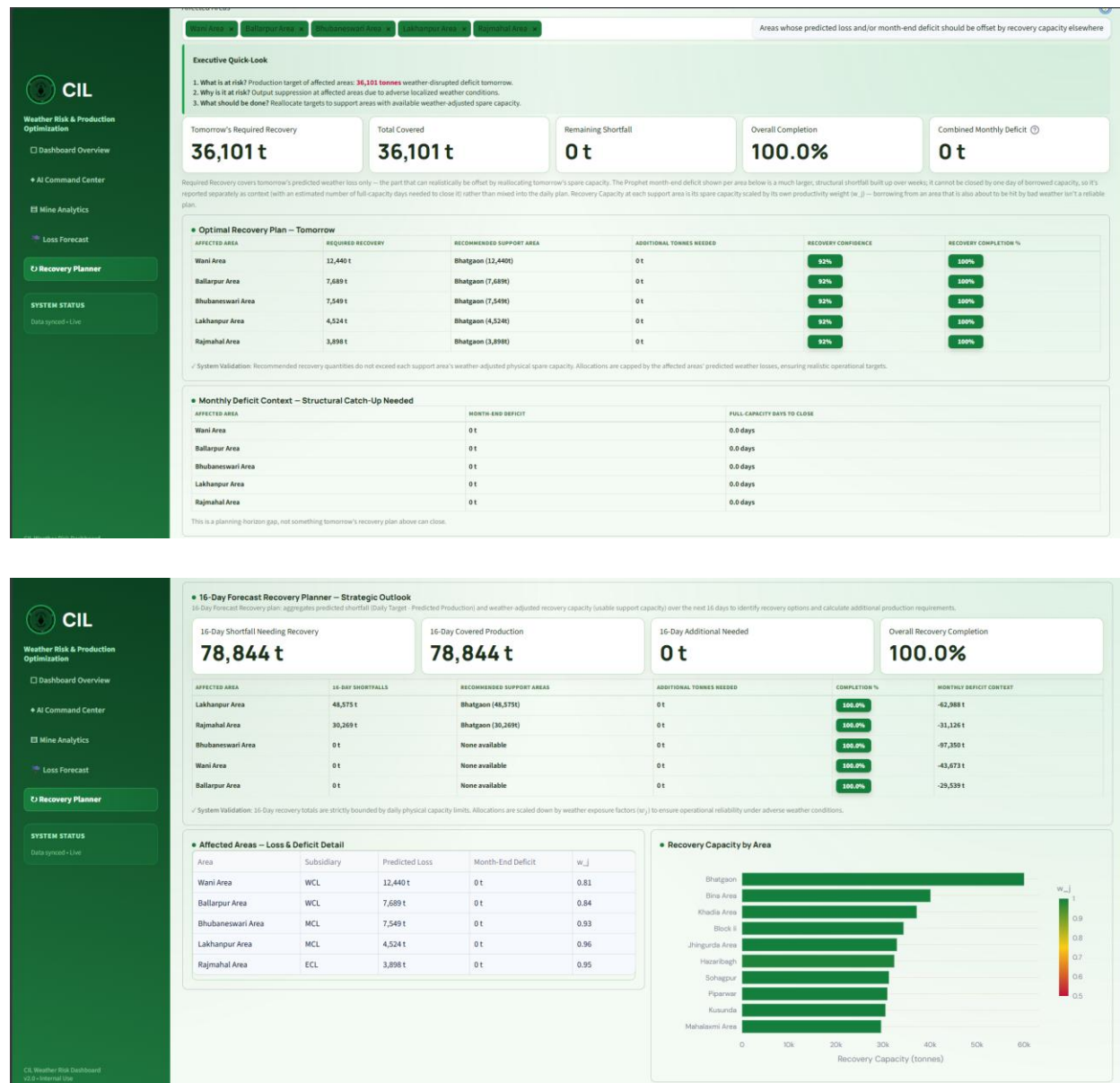
3. Mine Analytics

This page allows users to explore production and weather trends for individual mines or areas over time. for example, comparing current output to previous fiscal years, or seeing how rainfall patterns have historically tracked with production dips. It is the more exploratory, drill-down page for users who want to understand a specific mine in depth rather than the whole company at once.

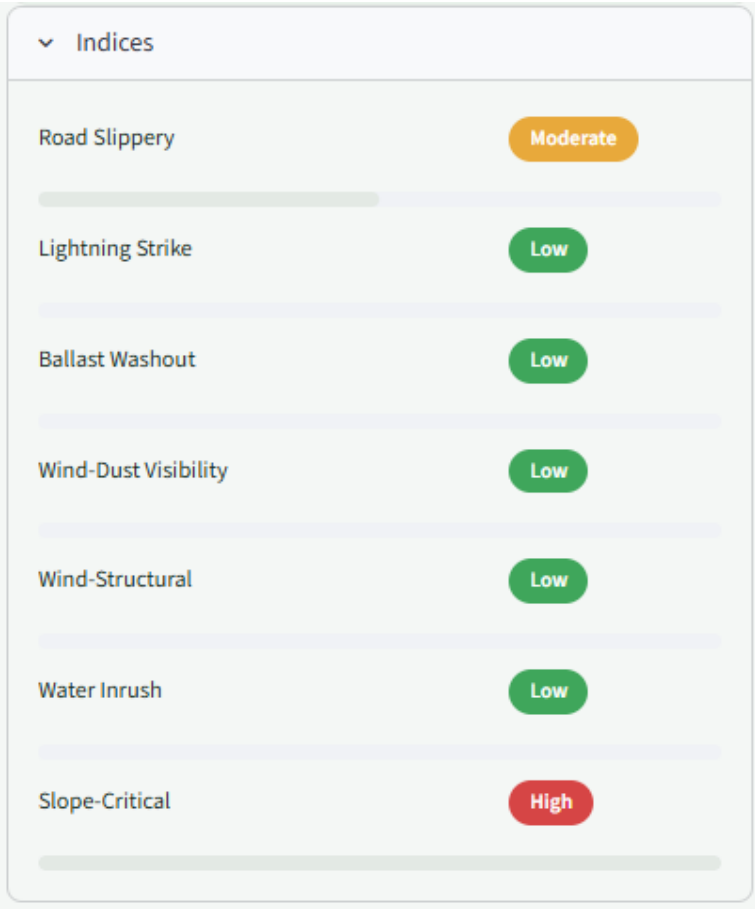
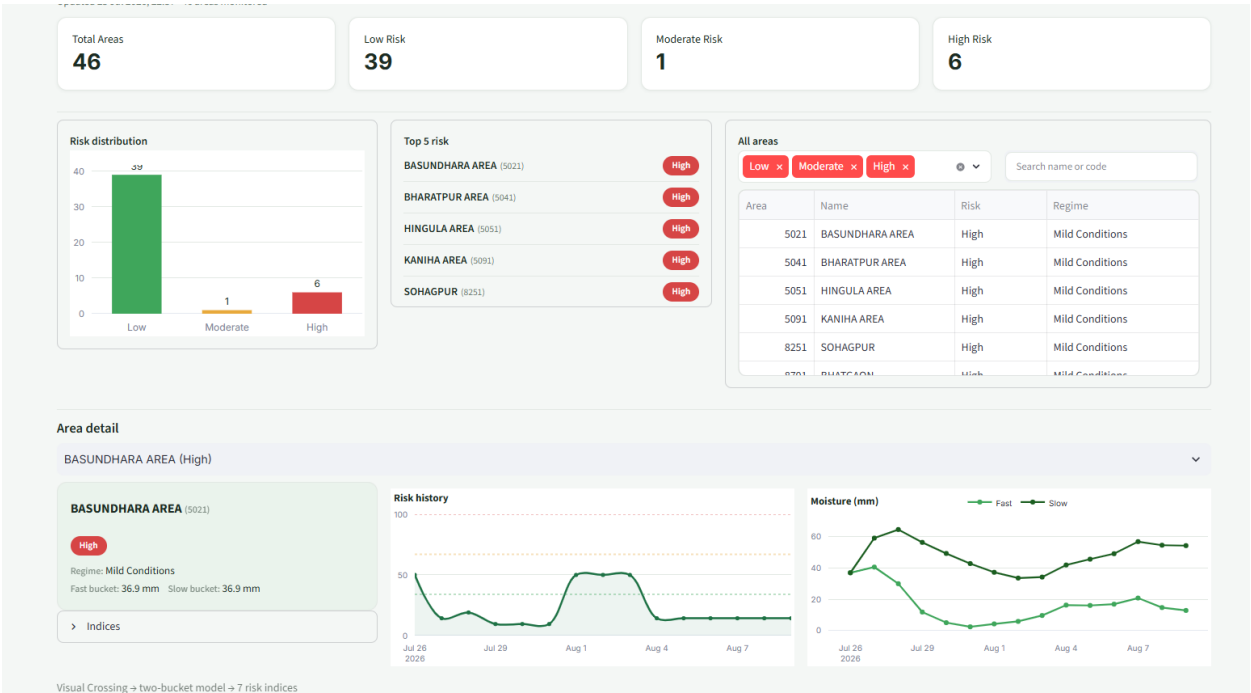


4. Recovery Planner

Weather-related losses are rarely permanent, production can often be partially recovered once conditions improve, by adjusting schedules, reallocating resources, or increasing effort at unaffected sites. This page turns forecasted losses into a recovery planner, showing how much of the expected shortfall could realistically be recovered given weather-adjusted capacity, and what a balanced week of operations could look like across areas. Answers 'here is how much of that you could get back, and how.' It effectively acts as a weekly planning aid: instead of every area independently deciding how to respond to weather disruption, the page provides a broad view of where spare capacity exists and where it might sensibly be redirected.



5. Risk Forecast



6. Dashboard Pages at a Glance

The table below summarizes the six pages described above, along with their primary audience and the main question each one answers.

Page	Primary Audience	Main Question It Answers
AI Command Centre	Executives, senior managers	Where are the problems today, across the whole company?
Loss Forecast	Area managers, planners	How much production is at risk, and how confident is that estimate?
Mine Analytics	Analysts, area managers	How has this specific mine performed and behaved historically?
Recovery Planner	Operations coordinators	What can we actually do to offset the expected loss?

7. Design Philosophy

Two ideas guided the visual design of the dashboard. First, everything needed to be understandable without special training clear colour coding (e.g. green for low risk, red for high risk), plain-language labels, and an uncluttered layout so an employee is not overwhelmed. Second, the dashboard needed to fit comfortably on a single standard screen for its main views, minimizing the need to scroll so that the 'headline picture' is always immediately visible. A consistent colour theme and shared visual style were applied across all pages so that the platform feels consistent.

8. A Typical User Journey

To make the dashboard's purpose concrete, consider how a subsidiary-level coordinator might use it on a Monday morning. They start on the AI Command Centre to get a company-wide snapshot and notice that two areas under their subsidiary are flagged as elevated risk. They click through to the Loss Forecast page, filter to their subsidiary, and see the specific forecasted loss range for each of those two areas over the coming week, along with the confidence band around that estimate. Wanting more context, they move to Mine Analytics to see how these two areas have historically responded to similar rainfall patterns in past fiscal years. Finally, they open the Recovery Planner, which suggests how the expected shortfall could be partially offset by shifting some capacity to unaffected sites for the week. Within a few minutes and without leaving the browser, the coordinator has gone from 'something might be wrong' to 'here is a specific, data-backed plan.'

9. Consistency Across Subsidiaries

Because CIL's seven subsidiaries historically operated with their own local reporting habits, one underrated benefit of the dashboard is standardization: every area, regardless of which subsidiary it belongs to, is measured using the same risk indicators, the same forecasting approach, and the same visual language. This means a manager comparing an area in ECL against an area in SECL is looking at genuinely comparable numbers, rather than numbers that were calculated or reported slightly differently by each subsidiary's own local process.

10. Value Delivered by the Dashboard

- Brings together production and weather data that would otherwise sit in separate spreadsheets and reports.
- Converts raw numbers into risk indicators that are immediately actionable (e.g. 'this area is high risk this week').
- Gives forward-looking forecasts, not just historical reporting, so managers can act before losses occur rather than after.
- Turns predictions into concrete recovery suggestions, closing the loop from 'what will happen' to 'what should we do about it'.
- Provides different levels of detail for different audiences — from a one-screen executive summary to mine-level deep dives.

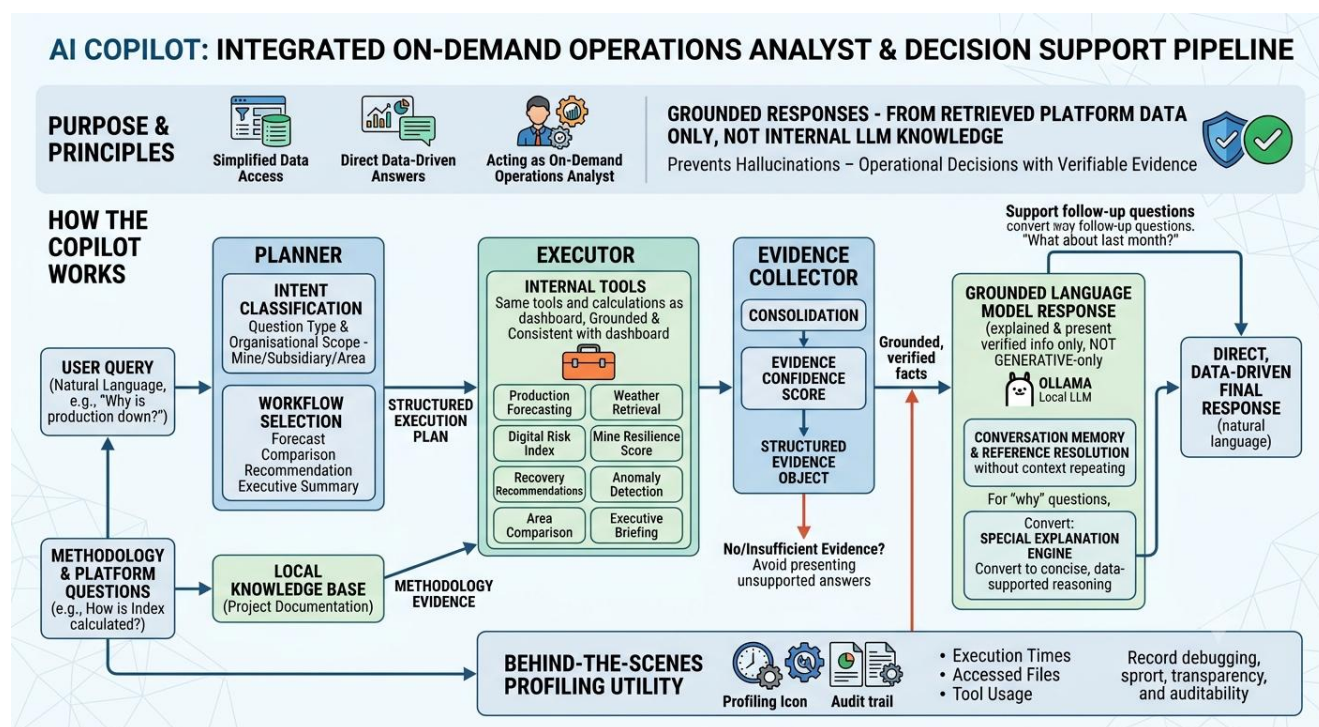
Chatbot

1. Purpose of the Copilot

The dashboard provides extensive operational insights but still requires users to navigate multiple pages, filters, and charts. The AI Copilot simplifies this process by allowing users to ask questions in natural language, such as *"Which areas are at high risk this week?"* or *"Why is production down at this mine?"*, and receive direct, data-driven answers. It acts as an on-demand operations analyst integrated into the platform.

A key design principle is that the Copilot is grounded rather than generative-only. Every response is based on real data retrieved from the platform's datasets, forecasting outputs, and documentation rather than the language model's internal knowledge. This prevents hallucinated values and ensures that operational decisions are supported by verifiable evidence.

How the Copilot Works



- **Planner:** Interprets the user's query by identifying the intent, scope, entity (mine/subsidiary), and workflow (e.g., Forecast, Comparison, Recommendation, Executive, or Methodology), then creates a structured execution plan.
- **Executor:** Runs only the required tools, such as production forecasting, weather retrieval, Digital Risk Index, Mine Resilience Score, recovery recommendations, anomaly detection, area comparison, and executive briefing.
- **Conversation Memory:** Maintains context and resolves follow-up questions (e.g., *"What about last month?"*) without requiring the user to repeat previous information.

- **Evidence Collection:** Consolidates outputs from all executed tools into a structured evidence object with a confidence score. Only evidence-backed information is used in responses.
- **Knowledge Base:** Answers methodology-related questions (e.g., forecasting or risk score calculation) using a dedicated local documentation repository rather than operational data.
- **Language Model & Explanation Engine:** A locally hosted **Ollama** model converts the collected evidence into natural-language responses, while a dedicated explanation engine generates clear, data-grounded explanations for "why" questions.
- **Monitoring & Debugging:** A profiling utility records execution time, tool usage, and accessed files, ensuring the pipeline remains transparent and auditable.

2. Workflow Types Recognized by the Planner

The Planner classifies every incoming question into one of several workflow types, each of which determines which tools are run and how the final answer should be formatted. The main workflow types are summarized below.

Workflow	Example Question	What It Triggers
Forecast	"What is next week's loss forecast for this area?"	Forecast and weather tools for the resolved mine
Ranking	"Which areas are highest risk right now?"	Comparison across areas, sorted by risk
Explanation	"Why is this area high risk?"	Full evidence collection plus the Explanation Engine
Methodology	"How is the Digital Risk Index calculated?"	Knowledge base lookup, not live data
Comparison	"Compare Area A and Area B"	Tools run for both entities, then merged
Executive	"Give me a company-wide summary"	Executive brief generation, global scope
Recommendation	"What should we do about this shortfall?"	Recovery and resilience tools
Validation	"How accurate have past forecasts been?"	Synthetic validation data lookup

 **Operations Console**

 Ask Copilot

 Explain This

 Daily Briefing

 Operations Analyst briefing terminal active. Ask me about mine weather risks, comparisons, recovery actions, or target deficits.

 which mine is at least risk in WCL



CIL AI Operations Copilot

• Operations Ranking: Digital Risk Index (DRI)

Rank	Mine Area	Subsidiary	Value	Risk Band

 Clear History

 Quick Summary

Ask Copilot...



 **Operations Console**

 Ask Copilot

 Explain This

 Daily Briefing

• **Operations Ranking: Digital Risk Index (DRI)**

Rank	Mine Area	Subsidiary	Value	Risk Band
1	Chandrapur Area	WCL	1.5	Low
2	Wani North Area	WCL	2.9	Low
3	Nagpur Area	WCL	4.0	Low
4	Umrer Area	WCL	4.2	Low
5	Majri Area	WCL	7.6	Moderate

 Clear History

 Quick Summary

Ask Copilot...



Page 51

 **Operations Console**

 Ask Copilot

 Explain This

 Daily Briefing

2	Wani North Area	WCL	2.9	Low
3	Nagpur Area	WCL	4.0	Low
4	Umrer Area	WCL	4.2	Low
5	Majri Area	WCL	7.6	Moderate

The provided evidence indicates that the ranking of mines in WCL based on DRI is already sorted in ascending order. Therefore, the mine with the highest risk category is actually the one with the lowest value, which is Chandrapur Area with a DRI value of 1.5 and categorized as "Low" risk.

 Clear History

 Quick Summary

Ask Copilot...



Operations Console

Ask Copilot

Explain This

Daily Briefing

Select a mine and metric to get a grounded explanation.

Select Mine Area

Gevra

Select Metric/Chart to Explain

Digital Risk Index (DRI)

Explain Metric

Grounded Explanation for Gevra:

CIL AI Operations Copilot

The Digital Risk Index (DRI) for Gevra mine is 1.2, categorized as Low risk. This value is derived from various evidence sources, including DRI, forecast, weather, resilience, validation, recovery, and deficit metrics.

A closer examination of the evidence reveals that the planned target for Gevra mine is 259230.0 units, with a forecast loss of 1061.2 units, representing a 0.409366199899703% loss. The weather productivity weight factor (w_j) is 1.0, indicating no impact from weather conditions on production. Rainfall and wind speeds are within acceptable limits, with rainfall at 4.4 mm and wind speed at 21.4 km/h. Waterlogging is minimal, with consecutive rainy days being zero.

The resilience score for Gevra mine stands at 70.7%, indicating a moderate level of resilience to risks. The forecast confidence is high at 79.8%. Notably, the recovery overall percentage is 100.0%, suggesting that any losses incurred have been fully recovered.

3. Data Sources and Reliability

The Copilot retrieves information directly from the same data sources as the dashboard, including historical production records, weather data, forecast outputs, the Digital Risk Index, and the Mine Resilience Score. This ensures that both the dashboard and the Copilot always present consistent information. To maintain reliability, every response is generated only from verified evidence. When data is incomplete or confidence is low, the system indicates this instead of producing unsupported answers. The language model is further constrained to use only the verified evidence supplied by the pipeline, preventing fabricated values or unsupported conclusions.

4. Example Workflow

For a query such as "Why is Area X at high risk this week?", the Planner identifies the request as an explanation for a specific mining area, the Executor retrieves the relevant forecast, weather, and risk data, and the Evidence Collector combines the results with a confidence score. The Explanation Engine then generates a data-supported explanation, while the language model converts it into a clear natural-language response. The final answer is therefore based entirely on the same information displayed in the dashboard.

5. Benefits and Future Scope

The proposed architecture improves reliability, consistency, context awareness, and auditability by ensuring every response is evidence-based and traceable to the underlying data and tools. Current limitations include the use of a locally hosted language model, limited conversation memory, and dependence on the availability of up-to-date operational data. Future enhancements include expanding the methodology knowledge base, adding voice and mobile support, and enabling proactive alerts that automatically notify users when monitored mining areas exceed predefined risk thresholds.

Conclusion

This internship project demonstrates the successful application of data science, machine learning, and weather analytics to address two important operational challenges faced by Coal India Limited: weather-induced production losses and weather-related accident risks. By combining historical weather observations, operational production data, synthetic data generation, and predictive modelling, the project developed a comprehensive decision-support platform capable of providing both historical analysis and forward-looking operational insights.

The production forecasting framework successfully captures area-specific relationships between weather conditions and mining performance through independent LightGBM models. The high predictive accuracy obtained during validation indicates that weather conditions, historical production behaviour, and engineered operational features provide strong explanatory power for production loss prediction. The incorporation of synthetic historical data also demonstrates a practical approach for addressing limited data availability while preserving realistic operational behaviour.

Alongside production forecasting, the rule-based accident risk framework provides an interpretable method for estimating weather-related safety risks across mining areas. Rather than relying solely on historical accident records, the framework combines domain knowledge with engineered weather indicators to generate actionable hazard assessments, enabling preventive planning before hazardous conditions develop.

The Streamlit dashboard integrates forecasting, visualization, recovery planning, and conversational AI into a unified operational platform. The addition of an AI-powered copilot further improves accessibility by allowing users to retrieve insights and explanations through natural language while ensuring that responses remain grounded in verified operational data.

Although the current framework focuses primarily on weather-related operational impacts, it provides a strong foundation for future expansion. Incorporating additional operational variables such as equipment availability, maintenance schedules, geological conditions, transportation constraints, IoT sensor data, and real-time mine monitoring could further improve forecasting accuracy and operational decision-making. Similarly, future versions of the accident risk framework could incorporate detailed geotechnical, hydrological, and terrain information to strengthen hazard estimation.

Overall, the project demonstrates that integrating machine learning, explainable artificial intelligence, weather forecasting, and interactive analytics can significantly enhance operational planning, improve resource allocation, reduce production uncertainty, and strengthen safety management within large-scale mining organisations. The developed framework provides a scalable foundation that can be extended to support more intelligent, data-driven mining operations across Coal India Limited and similar mining enterprises.

Future Enhancements

Future Enhancements for Production Loss Prediction

The current production forecasting model primarily estimates weather-induced production losses. Future versions of the model can be improved by incorporating additional operational, logistical, and equipment-related data, including:

- **Equipment Health Monitoring:** Integrate equipment availability, breakdown logs, maintenance schedules, and sensor data to account for production losses caused by machinery failures.
- **Fleet and Vehicle Telematics:** Incorporate haul truck GPS data, vehicle utilization, travel time, fuel consumption, loading cycles, and traffic congestion within mines to better estimate operational delays.
- **Workforce and Shift Information:** Include labour availability, absenteeism, shift schedules, overtime, and workforce productivity to capture human factors affecting production.
- **Explosive and Blasting Data:** Incorporate blasting schedules, explosive consumption, and blast success rates, as blasting delays often have a direct impact on production.
- **Coal Quality and Geological Characteristics:** Include seam thickness, stripping ratio, overburden characteristics, geological complexity, coal grade, and pit conditions to account for mine-specific productivity differences.
- **Stockpile and Dispatch Information:** Use stockpile levels, railway rake availability, siding congestion, dispatch capacity, and customer demand to better model the interaction between production and evacuation.
- **Power Supply and Infrastructure Data:** Incorporate power outages, conveyor belt availability, crusher utilization, and other infrastructure-related disruptions that affect production.
- **Real-Time IoT Sensors:** Use IoT devices to monitor equipment status, water levels, pit conditions, vibration, dust concentration, and environmental conditions for continuous model updates.
- **Satellite and Remote Sensing Data:** Integrate satellite-derived rainfall, flood extent, land surface temperature, and terrain information to improve spatial understanding of weather impacts.
- **Advanced Time-Series and Spatial Models:** Explore transformer-based forecasting models, graph neural networks, and spatio-temporal learning approaches to capture interactions between neighbouring mining areas and evolving weather systems.
- **Dynamic Target Planning:** Instead of assuming fixed production targets, future models can generate adaptive daily targets based on equipment availability, workforce capacity, planned maintenance, and forecast weather conditions.
- **Continuous Model Learning:** Develop an automated pipeline that retrains models periodically using newly available operational data, enabling the system to adapt to changing mine conditions and improve prediction accuracy over time.

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